Prometheus: Contextual Computational Primitives with Continuous State and Content Memory

Prometheus Research Project

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Abstract

How can a learning system acquire reusable computational operations while retaining earlier capabilities at acceptable resource cost? We investigate Prometheus, a trainable Block combining continuous state, low-rank nonlinear primitives, contextual calls and content-addressed memory. Encoders, operations, calling, writes and readout learn jointly from task feedback. Symbolic and continuous-input adapters enter the same core. Soft and hard-call execution are measured separately, and earlier topological constructions retain their own evidence boundaries. Fixed-budget development covers contextual programs, audio, vision and causal next-character prediction. In a fourth-stage two-address task, a 28,381-parameter dynamic Block reaches 56.64% IID accuracy, compared with 62.89% for a 25,616-parameter GRU, while older abilities decline. A frozen-state intervention reduces VALID accuracy from 58.20% to 12.5% when memory reads are removed; a joint key–value permutation preserves predictions. These results establish memory dependence, not a learning advantage. In a 614,400-label character experiment, dynamic-Block perplexity is 10.36, versus 10.53 for the tested Transformer, 8.66 for GRU and 7.17 for official Mamba2. Actual parameters, training and evaluation time, cold compilation, replay, optimizer state, inference and storage are recorded separately. Same-weight execution optimization reduces dispatch overhead, but the conventional architectures remain faster in the completed audit. These single-seed results identify acquisition, retention and computational bottlenecks rather than broad architectural superiority. A complete Primary architecture remains unfrozen. Ongoing controlled revisions target the learning and execution mechanisms exposed by these measurements.

1 Introduction

How can a learning system construct, compose, and improve its own internal computational mechanisms? Neural architectures already learn representations, select information, implement memory, and organize reusable computation. Attention, for example, learns input-dependent interactions through shared projections and weighted aggregation [5]. The question is how computational capabilities can accumulate under continued learning: which internal operations are acquired, how later tasks reuse them, and what must be preserved when the parameters implementing those operations change. An adequate answer should expose the executed computation and its learning history, together with the resources required to retain it.

This problem has two coupled scales. At the implementation scale, a shared parameter update changes representations, operations, and their invocation. At the semantic scale, a capability is a complete input–computation–output relation. Preserving an isolated operator is insufficient if its old inputs are subsequently encoded differently or routed elsewhere. Conversely, preserving all implementation details can prevent useful adaptation or require separate capacity for each task. Dynamically expandable networks illustrate that selective sharing, structural growth, and protection can be organized jointly [6]. Their existence motivates a precise comparison of learning organizations under shared information and resource conditions.

We investigate *Prometheus*, an executable Block with a continuous internal state, learned nonlinear residual primitives, a contextual caller, and content-addressed memory. The caller reads the current state, input and memory content to select or mix primitive updates. After those calls, learned addresses and gates revise the memory's keys and values. A task supplies observed inputs and a terminal label; gradients jointly train representation, operations, invocation, memory and readout. No task-specific primitive table or correct calling sequence is installed by the experimenter.

The architecture makes two interfaces explicit. First, symbolic embeddings and modality-specific projections enter the same core, permitting direct tests of a common computation on finite-state programs, text prefixes, audio and image features. Second, soft mixture execution and hard-call execution use the same parameters but follow distinct trajectories. Hard calls retain a continuous state and readout. Their quality and costs require separate measurement; replacing a mixture with its largest component is an empirical change to the computation.

This continuous design follows measured limitations of earlier categorical prototypes, including weak terminal credit and collapsed calling behavior. Residual state updates and learned memory are concrete architectural responses, whose effects can be tested against static calling, expanded state, recurrent networks, Transformers and applicable state-space models. The component operations are established mechanisms. The scientific target is an important acquisition–retention–resource benefit from their organization, with clear causal attribution. Historical monodromy-based protection remains an optional research mechanism documented with its original assumptions and controls.

Current evidence consists of bounded development runs. A contextual sequence experiment demonstrates learning with both execution interfaces; sampled character prediction tests length changes and parameter-matched controls; initial audio and image diagnostics exercise the continuous-input interface. The results are mixed and have not yet established continuing capability growth or a broad competitive advantage. This manuscript therefore connects an implemented, inspectable architecture to a prospective continual-learning evaluation. A validated Primary freeze and subsequent retraining separate exploratory changes from the eventual empirical claims.

2 Related Work

Context-dependent neural computation. Transformers learn input-dependent interactions through attention and combine them with shared feedforward transformations [5]. Prometheus also uses task-driven neural representations and weighted content selection. Its caller chooses among low-rank nonlinear residual functions; its externalized key–value state supplies a learned read vector to those functions. The static-caller control retains the same contextual primitives and memory, separating the effect of context-dependent invocation from the presence of a recurrent memory system. This comparison can identify an effect of the caller without treating memory, function composition or expert choice as new capabilities in themselves.

Recurrent and selective state-space models provide additional conventional organizations of context. Gu and Dao's selective SSM makes step sizes and write/read vectors depend on the current input, controlling how information enters and propagates through state.[1] Its state parameterization and recurrence differ from the explicit primitive mixture considered here. Comparisons must also distinguish an author's optimized implementation from a reference adaptation: mathematical recurrence alone does not reproduce fused scan, recomputation, or kernel performance.

Continual organization. DEN selectively retrains relevant subnetworks, conditionally expands capacity, and uses splitting and timestamped inference to manage semantic drift [6]. Replay, freezing and modular capacity are likewise concrete comparison conditions in our protocol. The present continuous Block requires its own retention measurements during shared learning. Parameters, stored examples, auxiliary states, maintenance and execution must all enter comparisons; equivalent terminal supervision does not automatically equate total work or attained capability.

Discrete mechanisms and topological protection. Earlier Prometheus constructions used class-boundary feedback for loop operations. Their derivative decomposition follows established stochastic computation-graph and discontinuous-gradient analyses [4, 3]; it is distinct from ordinary differentiable training of the current residual functions. Monodromy supplies a conditional operation-invariance mechanism, retained in the appendix as an optional component. Kitaev's braiding construction concerns physical topological quantum protection [2]; classical loop simulation does not inherit that hardware guarantee. Bennett's reversible simulation explicitly retains and later cleans auxiliary information [1]. It informs historical

[1] Original mechanism: Algorithm2 and Sections3.2–3.4 of `https://arxiv.org/abs/2312.00752`.

workspace analyses, whereas the current nonlinear state and memory overwrite rules directly allow irreversible updates. Each organization needs its own learning, retention and resource evidence.

3 Prometheus Architecture

Prometheus investigates a trainable computational Block whose internal operations, calls, and memory are learned together from task feedback. The current development architecture uses a continuous state, a bank of nonlinear residual primitives, and content-addressed memory. This revises the earlier categorical and loop-based implementations without transferring their protection guarantees to the new model. Its ingredients are established neural mechanisms; the question is whether their particular organization supports useful acquisition, composition, and retention at competitive total cost.

3.1 State, inputs, and executable primitives

Let $h_{t,s} \in \mathbb{R}^H$ denote the state before call slot s of input t. Memory contains keys $\mathsf{K}_t \in \mathbb{R}^{M\times Q}$ and values $\mathsf{V}_t \in \mathbb{R}^{M\times H}$. An observed initial-state identifier selects a learned embedding $h_{1,0} = E_\theta[x]$. Keys begin at a learned, task-independent matrix and values at zero for each example. There is no persistent memory across examples in this implementation.

The raw input feature $u_t \in \mathbb{R}^T$ is a token embedding for symbolic inputs or the output of a learned modality adapter. A learned slot embedding gives $x_{t,s} = u_t + e_s$. All slots reuse the same primitive bank and caller. During a token, memory is fixed while the state changes. Each call computes

$$r_{t,s} = \mathrm{softmax}\Big(q_\theta(h_{t,s}, x_{t,s})\mathsf{K}_t^\top/\sqrt{Q}\Big)\,\mathsf{V}_t, \tag{1}$$

$$\alpha_{t,s} = \mathrm{softmax}\, f_\theta(h_{t,s}, x_{t,s}, r_{t,s}), \tag{2}$$

$$z_{t,s} = [h_{t,s}; x_{t,s}; r_{t,s}], \tag{3}$$

$$\delta_{t,s,k} = U_k\,\mathrm{GELU}(D_k z_{t,s} + b_k), \tag{4}$$

$$h_{t,s+1} = \mathrm{LayerNorm}\left(h_{t,s} + \sum_{k=1}^{K} \alpha_{t,s,k}\delta_{t,s,k}\right). \tag{5}$$

Here $D_k \in \mathbb{R}^{R\times(2H+T)}$ and $U_k \in \mathbb{R}^{H\times R}$. Each primitive is an executable nonlinear state update with its own learned parameters. The bank has no supplied task-specific operation or explicit identity choice. A residual path replaces repeated row-stochastic state propagation; neither normalization nor low rank implies invertibility or exact preservation of information.

3.2 Content memory and composition

After the last call, write features concatenate the final state, raw input, and last read vector: $\bar{z}_t = [h_{t,S}; u_t; r_{t,S-1}]$. Learned projections produce an address, gate, candidate key and candidate value:

$$a_t = \mathrm{softmax}(w_\theta(\bar{z}_t)\mathsf{K}_t^\top/\sqrt{Q}), \qquad g_t = \sigma(g_\theta(\bar{z}_t)), \tag{6}$$

$$\widehat{k}_t = \tanh(k_\theta(\bar{z}_t)), \qquad \widehat{v}_t = v_\theta(\bar{z}_t). \tag{7}$$

With $\beta_{tj} = g_t a_{tj}$, each slot updates as

$$(\mathsf{K}_{t+1,j}, \mathsf{V}_{t+1,j}) = (1-\beta_{tj})(\mathsf{K}_{t,j}, \mathsf{V}_{t,j}) + \beta_{tj}(\widehat{k}_t, \widehat{v}_t). \tag{8}$$

The next token starts from $h_{t+1,0} = h_{t,S}$. Thus later calls depend on the actual intermediate computation and retained content. Order-sensitive composition follows from executing successive nonlinear updates; no commutativity constraint is imposed. The terminal distribution is $\mathrm{softmax}(W_R h_{L,S} + b_R)$.

The development configuration has $H = 64$, $T = Q = 16$, $M = 8$, $K = 4$, $R = 8$ and one call per token. All dimensions are explicit resources, rather than counts inferred from a hidden task generator. A static-caller control zeros only the state and memory-read inputs to f_θ; its primitives, memory, shapes and parameter count are unchanged. It therefore retains contextual computation and is stronger than a memoryless ablation.

3.3 A common input interface

Symbolic lookup and externally embedded inputs enter the same execution function. Audio features or image patches require a learned projection to $\mathbb{R}^T$, whose parameters and work are charged separately. No modality adapter supplies operations, calls, intermediate targets, or task identities. This common interface permits testing the same core across tasks; it does not by itself establish transfer or continual retention. The complete Primary architecture remains unfrozen, and present measurements are separately identified development evidence.

4 Learning and Retention

4.1 Joint operation formation from terminal feedback

For observed examples $(x, u_{1:L}, y)$, the current learner minimizes terminal negative log likelihood,

$$\mathcal{L}(\theta) = -\frac{1}{B}\sum_{i=1}^{B} \log p_\theta(y_i \mid x_i, u_{i,1:L_i}). \tag{9}$$

The initial representation, primitive matrices, caller, memory read/write projections and readout all receive gradients through the executed sequence. The learner receives neither correct calls nor intermediate states. Primitive formation here means changes to executable learned state updates, rather than assignment of examples to an already supplied answer table. It does not imply that a primitive becomes a separately named or independently interpretable algorithm. Such behavior requires execution and intervention evidence.

Soft execution evaluates every primitive and combines its residual according to the learned call probabilities. This supplies terminal credit to the whole bank. Backpropagation through time also differentiates the learned content memory; state and memory reset for each example. The continuous development runs use soft training. Their optimization budget, sampled minibatches and validation protocol are fixed independently of test scores. Learning curves and failed predecessors are retained rather than replacing low results with an assumed successful configuration.

4.2 Soft and hard-call execution

Hard-call execution selects $\widehat{k}_{t,s} = \arg\max_k \alpha_{t,s,k}$ and evaluates only $\delta_{t,s,\widehat{k}_{t,s}}$. Encoding, state, memory, primitive functions and the probabilistic readout remain continuous. We call this interface *native* to distinguish it from mixture execution; it is not a fully discrete program or a one-hot classifier. Both interfaces use the same saved weights, and both yield normalized terminal distributions.

An optional straight-through interface uses hard calls in the forward pass and a dense soft-mixture surrogate in the backward pass. Writing $\delta^{\text{soft}} = \sum_k \alpha_k \delta_k$ and using sg for stop-gradient, its residual is

$$\delta^{\text{ST}} = \text{sg}(\delta_{\widehat{k}}) + \delta^{\text{soft}} - \text{sg}(\delta^{\text{soft}}). \tag{10}$$

This is a biased surrogate for a discontinuous decision, not its exact gradient. Soft and native trajectories can diverge because subsequent calls and memory writes depend on their resulting states. Consequently, likelihood, accuracy and inference cost must be measured separately; correctness of soft execution does not certify a hard-call program.

4.3 Retention as an empirical obligation

Shared updates may change old encodings, primitive behavior, calls, memory access and readout. The present continuous Block has no built-in theorem protecting their composition. Its continual-learning policy must therefore be evaluated against replay, freezing, modular adaptation and other applicable controls with the same information. Extra examples, snapshots, auxiliary state and maintenance are counted. Finite current-task success cannot replace measurements during and after later learning.

Earlier implementations supply a distinct optional protection mechanism: homotopic deformations of a based loop preserve its monodromy action, while additional finite closure checks protect learned task bindings.

Those constructions and their counterexamples remain in Appendices B and C. Their certificates apply to the specified finite execution models, not to the unconstrained nonlinear primitives above. Monodromy is not required for the current Primary candidate; its later inclusion would need an explicit integration and a measured benefit beyond strong ordinary controls. Neither this optional component nor an independent new learning rule is presumed necessary for a useful architecture.

5 Theoretical Analysis

We analyze the implemented resource structure, separating trainable weights, transient state and execution. These are conditional accounting identities and explicit work models, not claims of superior complexity or protection.

Parameters and working state. Let N be the output/initial-state cardinality, V the input vocabulary, H the state width, T the token width, Q the key width, M memory slots, K primitives, R primitive rank, C caller width and S calls per token. Set $F = 2H + T$. Including biases, the current Block has

$$\begin{aligned} P = {} & NH + VT + ST + MQ + \big((H+T)Q + Q\big) \\ & + (FC + C + CK + K) + K(RF + R + HR) + 2H \\ & + 2(FQ + Q) + (FH + H) + (F + 1) + (HN + N). \end{aligned} \tag{11}$$

The terms account respectively for learned interfaces and initial keys, queries, calls, primitives, normalization, writes and readout. A modality projection adds its own parameters. Increasing sequence length adds no model weights, while increasing H, M, K or rank changes both capacity and cost.

The recurrent payload for one example is

$$C_{\text{state}} = H + M(Q + H) \tag{12}$$

scalars. For the stated development dimensions this is 704 floats. It excludes temporary tensors, gradients, optimizer moments and saved training activations. A streaming implementation can retain this state and process a new input; the current forward API instead resets per example. Current character experiments recompute each entire observed prefix. Their timings therefore do not establish a streaming or autoregressive caching advantage.

Forward work and training memory. For B examples and maximum supplied length L, dense primitive projections contribute at most

$$W_{\text{primitive}} = BLSKR(F + H) \tag{13}$$

multiply–accumulate operations in soft execution, versus $BLSR(F + H)$ for actual selected-expert evaluation. This conditional saving does not equate their predictions. Memory reads additionally require query projection and $O(BLSM(Q + H))$ attention products; writes and the caller have their own projection and update terms. Source counters report these components separately. They are forward MAC proxies, not measured training FLOPs: they exclude normalization, activations, indexing, backward and backend workspace. Actual wall time and allocator peaks remain necessary measurements.

Backpropagation through a prefix retains intermediate states and operator activations. Its memory consequently grows with unrolled length even though the inference carry is fixed. Parameter storage, recurrent payload, training memory, checkpoint bytes and inference latency are distinct resources.

Composition and protection boundaries. Repeated calls define an explicit recurrent composition. Low-rank residual updates, LayerNorm and content addressing do not guarantee exact inversion, arbitrary memory recovery, length extrapolation, or preservation after shared learning. Likewise, the finite stochastic-path certificate of the earlier categorical model is not a certificate for this context-dependent continuous model. Historical topological preservation and reversible-workspace results retain their original assumptions in the appendix. The current architecture must establish useful acquisition and retention through the matched empirical comparisons of Section 6.

Model	Parameters	Task0	Task1	Task2	Train s
Continuous dynamic	28,365	82.81	90.23	91.99	92.21
Continuous static caller	28,365	81.45	94.92	94.14	79.12
GRU44	25,572	93.95	95.90	93.36	10.25
Transformer32	26,728	48.63	79.30	83.01	13.76

Table 1: Final IID accuracy (%) after three development phases, without replay. Train s sums actual gradient-step wall time, not total work or exclusive throughput. The static caller retains contextual primitives and memory.

Model	P	PPL64	Train s	Driver s	Forward G MAC
Dynamic	41,593	10.358	75.96	119.47	21.715
Static	41,593	10.306	75.24	118.74	21.715
Detached	41,593	10.243	74.07	114.41	21.715
Transformer36	39,664	10.533	6.75	12.26	29.816
GRU44	32,660	8.664	2.49	7.96	16.977
Official Mamba2	41,116	7.173	3.56	4.58	—

Table 2: Actual all-token language development. Driver time includes measured evaluation/checkpoint work but excludes process startup. MAC proxies count training forwards only, not backward or total FLOPs; all raw costs are retained.

6 Experiments

Our evaluation asks whether contextual computational primitives improve acquisition, retention, or resource use relative to strong conventional architectures. Topological components receive a separate causal test when included. The primary experimental object is the latest fully fused Prometheus architecture after general Block validation. Monodromy is an optional component selected by measured benefit; its independent advantage is not a prerequisite for the Primary freeze. Its core mechanism, code version, and evaluation configuration will be frozen before retraining across all five evaluation layers. No publication results for the to-be-frozen Primary are available at the present manuscript checkpoint. Earlier coordinate-renderer streams and small integrated diagnostics are development evidence in Appendices E and F; they are not pooled with Primary results or used to supply its performance or efficiency claims.

6.1 Observed development results

We first tested the same continuous core on three-stage contextual programs, sampled character prediction, and initial real-input recognition. These are fixed-budget, single-seed development measurements, not formal Primary results. The context stream exposes raw tokens and an initial state, with terminal labels but no task identifier, correct route, or intermediate state. Training lengths are 2–12; separate tests use lengths 24–48 and structured unseen compositions. Each phase uses 600 updates of 64 examples. Actual phase-0 weights, Adam state, and RNG were restored for later phases; phase0 was not retrained.

The dynamic model forgets part of task0 and does not uniquely outperform the static caller. Distribution checks are essential: a TRAIN-majority constant predictor scores 69.73% on task2's long test, equal to the dynamic model's 69.73%. Its structured-unseen scores are 25.78% and 25.98%, respectively. Thus high IID accuracy alone does not establish general memory computation.

All-token character development uses 300 updates of 32 length64 windows, 614,400 next-character labels per method. A causal-mask Transformer and GRU receive identical windows and batches. Future-token perturbations test actual causality before training. The 512 frozen TEST endpoints are paired across lengths64/128/256; length comparisons score the same final 64 labels. These are sampled-window character PPLs, not full-corpus or word-level PPLs.

With contexts of 64, 128, and 256 characters, dynamic common-suffix PPL is 10.358, 10.251, and 10.253; Transformer yields 10.533, 15.185, and 14.736, and GRU yields 8.664, 8.552, and 8.552. Stability under

Model	Speech acc.	Image acc.	Speech/image P	Train s (S/I)
Dynamic	29.77	42.10	32,280/40,793	97.51/66.28
Static caller	71.17	40.10	32,280/40,793	98.49/66.05
Detached	69.38	39.00	32,280/40,793	97.84/65.97
GRU64	76.25	44.60	54,819/59,556	1.60/1.92
Transformer64	76.48	49.90	105,379/110,116	7.92/7.85
CNN	59.77	44.30	35,683/38,852	.96/1.05

Table 3: Initial Speech Commands/CIFAR-100 development TEST accuracy (%). Each uses the first 10 official classes, 128 TRAIN examples per class, common features and 300 updates. Full 35/100-class heads are charged. This is not yet full-dataset or class-incremental validation. Training wall times reflect the original recorded possible concurrency, not isolated throughput.

longer inputs alone does not prove useful distant-memory use. The earlier terminal-prefix protocol and serial eager/hard-call inference audits remain separately reported in the appendix and are not pooled into this new protocol. They show no hard-call speedup. Cached streaming serving has not been established. The subsequently executed official Mamba2 baseline uses the same 300-update protocol and achieves common-suffix PPL 7.173/7.048/7.049 with 41,116 parameters. Its 3.56-second training and 4.58-second driver times exclude 82.67 seconds of measured cold compilation/priming; the entire coordinator launch takes 93.31 seconds and creates 55.19 MB of kernel cache. Its exact selective-SSD and backward FLOPs remain unmeasured. Earlier arms have their own recorded concurrency, so this is not an isolated-throughput ratio. These results provide no broad competitive advantage for the current Block.

The detached variant preserves the dynamic forward rule and parameter count, but stops only the caller-input gradients from state and memory. CPU autograd checks verify this intervention. It improves the poor speech optimization trajectory, while not improving the image result; it is not a general solution. Training and evaluation work, optimizer state, input caches, and failed versions are recorded separately. The current Python-unrolled Block is slower than the competitors despite smaller forward MAC proxies. Consequently these observations do not establish a broad capability or complete-resource advantage.

6.2 Questions and comparison conditions

We distinguish three questions: whether new computations are acquired from terminal feedback, whether later learning preserves their complete execution, and whether topology independently changes the acquisition–retention–resource tradeoff. Preservation includes input encoding, operation selection, composition, and readout, rather than operation maps alone. All methods receive the same permitted examples, terminal labels, stage order, and inference information. Teacher transitions, intermediate execution states, correct routes, and preconstructed task-specific primitives are excluded as free supervision. Discovery, installation, observation-table passes, and protection are charged.

We separately compare matched information, matched resource budgets, and matched attained quality. Equal update counts alone do not equate total learning work when a method also performs program discovery or verification. Accuracy–parameter, accuracy–training-cost, and accuracy–inference-cost frontiers expose these differences. Changes to input adapters and output heads are recorded and charged; a changed core learning rule constitutes a different architecture variant.

6.3 Tasks and continual streams

The finite computational streams expose observable initial states, token programs, public task context, and terminal outputs. Context-dependent token semantics make that public context available to every competitor. We disclose short-word coverage, repeated examples, and every additional label access. Both differentiable neural execution and native deployment are evaluated on new and old tasks. Intermediate old-task minima and their subsequent recovery are reported separately from stage-boundary retention.

The composition suite includes $AB \neq BA$, commutators, and unseen operation combinations. Training lengths are at most eight; tests at lengths 16, 32, and 64 are reported separately. Key–value tasks require writing, overwriting, reading, and delayed queries from terminal supervision. Generality tests cover merging

Layer	Task family	Principal measurement
I	Six-stage streams at $N = 8, 16, 32$	Acquisition, stage-boundary and within-stage retention, parameter growth
II	Noncommuting programs, long compositions, key–value memory	Order sensitivity, unseen composition, delayed state use
III	Reversible, nonreversible, continuous, variable-length computation	Generality of the same core Block
IV	Speech Commands v2 and Split CIFAR-100	No-task-ID class-incremental recognition and forgetting
V	Architecture competitors and component interventions	Competitive frontiers and independent mechanism attribution

Table 4: Five required empirical layers. Success on one does not replace measurements at another. The complete protocol is in Appendix E.

Setting	Low budget	Medium budget	High budget
Synthetic, updates/phase	B1:600	B2:2,400	B3:9,600
Real data, epochs/phase	10	50	200

Table 5: Prospective budget tiers available to every method. Discovery, maintenance, and configuration-development work are additional reported costs.

or erasing operations, continuous inputs/outputs, and variable-length execution. Reversible lifting, extra workspace, or external memory counts toward resources rather than making irreversibility free.

The following formal protocol remains prospective. Speech Commands v2 uses 35 classes in seven phases of five, the official train/validation/test split, and common 16-kHz preprocessing. Split CIFAR-100 uses 100 fine classes in ten phases of ten, the official 50,000 training and 10,000 test images, and a fixed class-stratified 10% training holdout for validation. All methods share preprocessing and augmentation. Both tasks recognize all seen classes without a task ID or class-group identifier at inference. Their Memory-0 and Memory-500 conditions permit zero or at most 500 stored old examples in total. Stored features, teacher outputs, and statistics are additionally disclosed. Main comparisons start from random initialization; pretraining is supplemental with equal access across methods.

6.4 Competitors and bounded development

The primary competing family is the Transformer: vanilla, experience replay, adapters, and replay plus adapters, at two prospectively selected scales. Continual-learning comparisons include DEN, PackNet, replay, and ER-ACE or a mature comparable class-incremental algorithm. Task-incremental PackNet is separated from no-ID class-incremental rankings. Mamba-2 participates in sequence/memory tests, Titans in applicable memory tasks, and HOPE/Nested Learning where a verifiable implementation supports the interface. Ordinary program learning, freezing, modular expansion, and constraint optimization provide additional matched-information controls; they do not replace neural architecture comparisons.

Each main family has at most 24 development configurations. Development seed 0 selects candidates using frozen validation criteria; the top three are checked with development seed 1. Configuration lock precedes final seeds 0, 1, 2, 3, 4. Test and OOD results never select hyperparameters, retries, stopping points, or configurations. Methods may use their appropriate optimizers and schedules. Extra-budget trajectories remain separately labeled supplemental configurations. Only localized, reproducible software, data, or training-logic defects qualify as bug fixes; ordinary tuning is development.

Configuration validity depends on checked implementation, fidelity of any claimed reproduction, information fairness, and compliance with the prospective protocol, never on an accuracy threshold. High, intermediate, and random-level scores are all valid outcomes under those conditions. A low-scoring adaptation supports conclusions about that adaptation and budget, not an entire published family. Unfaithful reproductions and untested implementations are explicitly identified; they are not made invalid by low performance.

Failed acquisition, all failed seeds, and additional resource use are retained without extending trials until a favorable result appears.

Learning curves distinguish failure to acquire a first task from forgetting after later tasks. Training and validation diagnostics examine representation limits, optimization, and credit assignment without using held-out tests to choose a repair. A substantive architecture or optimizer revision receives a new configuration identifier and preserves its predecessor's results. Formation before gradient training, maintenance after training, and admission are separate events, with their own label accesses and costs. This prevents installed programs or confidence sharpening from being misreported as capabilities first discovered by the gradient updates.

6.5 Attribution, costs, and statistical analysis

Same-architecture interventions remove reuse, sharing, state growth, old-capability protection, graph constraints, and native compilation separately. Topology-specific controls substitute ordinary constraints, disable cross-class loop learning, disable within-class safe deformation, and compare ordinary boundary optimization or discrete search. Information, certificates, and caches are shared when applicable. Benefits from compression, replay, ordinary closure, or confidence maintenance are not attributed to monodromy. A topology benefit requires a measured residual effect under these concrete controls.

We record total, trainable, and frozen parameters; actual model, optimizer, cache, replay, reference, and checkpoint storage; updates and sample accesses; training FLOPs and GPU time; discovery/search and rejected proposals; growth, maintenance, protection, verification, and reload. Controlled same-hardware inference reports latency p50/p95, throughput, and peak memory. Concurrent wall times are not ranked as isolated throughput, and overlapping timing/storage fields are not summed twice. Parameters per newly retained capability include the initial allocation and do not hide costs in nonparameter state.

Final accuracy, average incremental accuracy, forgetting, backward transfer, acquisition cost, and equal-quality resource use are reported over all five independent final seeds. We report means, standard deviations, 95% confidence intervals, and paired-seed differences; multiple primary tests receive multiplicity correction. Resumed fragments of one lineage are not independent replicates. Until these frozen-Primary measurements exist, this protocol specifies the evaluation rather than asserting an experimental advantage.

7 Discussion & Limitations

The current Block makes computational operations and their invocation explicit within a trainable continuous system. Its contextual sequence development run shows that terminal supervision can train useful nonlinear state updates and calling behavior. The same-size static-caller model is also strong. Because that control still gives the primitives and memory full contextual inputs, these observations do not isolate a benefit of dynamic calling. The useful next mechanistic question is which learned primitive functions, memory reads and invocation decisions account for the acquired computation, and whether those roles remain reusable when task demands change.

Parameter-matched character development makes the comparison more specific. After the same 300 updates, the 41,593-parameter dynamic Block has sampled perplexities 12.458, 12.468 and12.496 at contexts 32, 64 and 128, respectively. The static variant has 12.383, 12.452 and12.436. The 43,300-parameter Transformer has 12.803, 49.837 and48.409, while the 41,548-parameter GRU has approximately 10.090 across these contexts. Context 64 is the designated unseen-length split and context 128 the long split. These are 512 paired sampled positions per split, not whole-corpus or word-level perplexities. They show a stable contextual behavior for the tested Block but also a stronger recurrent control. The Transformer's long-prefix result characterizes this configuration and budget, not the whole architecture family. Matching parameters does not match training work, forward cost or transient memory.

Initial real-input diagnostics expose a further gap. With the first ten official classes, at most 128 training examples per class and 300 updates, the speech Block has approximately 29.77% soft and 22.19% hard-call accuracy, compared with 76.48% for the larger Transformer. The image Block obtains approximately 42.1% and 30.4%, respectively, compared with 49.9% for the Transformer. Their Block parameter counts are 32,280 and 40,793; the Transformer counts are 105,379 and 110,116. These static development tests exercise the

same core through learned adapters. They do not yet test full-class continual recognition, and the soft/hard gaps show that inference selection remains a material architectural choice.

The central remaining obligation is continual acquisition with complete retention and resource accounting. The continuous Block currently resets per-example memory, recomputes observed prefixes and has no general shared-update protection certificate. Earlier finite monodromy and binding-closure results retain their narrower scope in the appendix. Any integration must explain what operation or task relation is protected in the new state space and measure its benefit against ordinary constraints. An independent new primitive is not a prerequisite for an important empirical result; a broad reproducible advantage still requires a frozen architecture, strong appropriate controls, and the full evaluation rather than a favorable subset of development scores.

8 Conclusion

Prometheus investigates how learned operations, contextual invocation and content memory can organize executed computation. The current implementation uses continuous state and low-rank nonlinear residual primitives, jointly trained with representation, memory and readout from terminal labels. A shared core accepts symbolic or continuous features and supports separately measured soft and hard-call interfaces. Bounded sequence, sampled-character, audio and image development provides concrete learning and resource evidence, with mixed outcomes against recurrent and Transformer controls. It does not yet establish broad advantage or protected continual capability growth. Historical topological constructions and their negative controls remain available as distinct optional mechanisms. The next scientific claim must come from the complete architecture's acquisition, retention and cost under frozen fair comparisons. The present manuscript records that architecture and evidence boundary while the Primary design and formal evaluation remain in progress.

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A Continual acquisition and capability growth

Historical mechanism and Primary architecture boundary. This section records the earlier native/coordinate-renderer prototype as historical mechanism evidence and internal ablations. It is not the Primary Prometheus architecture for the publication benchmark. Primary Prometheus must be the latest fully fused architecture actually validated under TODO 3; its core mechanism, code version, and evaluation configuration must be frozen before all five benchmark layers are retrained. Historical streams cannot be pooled with those new Primary results or used to supply their parameter and cost claims. The limited D3 hybrid V2 diagnostic is subsequent development, not completed TODO 3 validation or a frozen Primary architecture.

The historical acquisition system learns finite recurrent computations from terminal examples, installs the inferred operations in neural parameters, and subsequently trains and maintains a differentiable calling interface. This is a hybrid identification–installation–gradient-learning procedure. Its successful synthetic executions establish a working accumulation mechanism within the tested task family; they do not yet establish that topology improves acquisition or parameter growth. In particular, an ordinary constrained implementation with the same formation and maintenance rules matches the measured results.

A.1 Historical representations, operations, and calling

A task example consists of an observable initial state $x \in \{1, \dots, N\}$, a token word $w = (a_1, \dots, a_L)$, a public context c, and a terminal label $y \in \{1, \dots, N\}$. The public context is necessary in these synthetic streams because identical tokens have different meanings in different tasks. It is not a hidden-state label. The learner does not receive target transition tables, intermediate execution states, or correct call routes.

The shared neural state is a row distribution $\alpha \in \mathbb{R}^{1\times D}$. Let $E \in \mathbb{R}^{N\times D}$ and $R \in \mathbb{R}^{D\times N}$ denote encoder and readout logits. At temperature $\tau > 0$, encoding and decoding use row softmaxes, denoted $\overline{E}$ and $\overline{R}$. A bank contains K learned operations and a provided identity operation $P_0 = I_D$. The latter is an execution substrate primitive, not a learned discovery. For token a in context c, a learned network produces S slot distributions $q_{c,a,s}(k)$, $k \in \{0, \dots, K\}$. Soft execution is

$$\alpha_0 = \overline{E}_{x,:}, \tag{14}$$

$$T_{c,a} = \prod_{s=1}^{S} \left(\sum_{k=0}^{K} q_{c,a,s}(k)\, \overline{P}_k \right), \tag{15}$$

$$\alpha_\ell = \alpha_{\ell-1} T_{c,a_\ell}, \qquad \widehat{p}(y \mid x, w, c) = (\alpha_L \overline{R})_y. \tag{16}$$

Products are ordered left to right in execution order because states are row vectors. Hard execution selects the encoder state, slot calls, operation matchings, and decoded label. For a destination permutation π_k, $(P_k)_{ij} = \mathbf{1}\{j = \pi_k(i)\}$; consequently $e_i P_k = e_{\pi_k(i)}$. The continuous monodromy realization executes the selected operation through path lifting. Its mathematical role and later coupling to learning are developed in the monodromy section.

The compact bank stores destination coordinates $\xi \in \mathbb{R}^{K\times D}$ and generates matching scores

$$W_{kij} = -8(j - \xi_{ki})^2. \tag{17}$$

Every permutation is representable by setting $\xi_{ki} = \pi_k(i)$. The persistent bank therefore contains KD learned scalars, although forming and differentiating its dense scores still incurs KD^2 temporary work and storage. The ordinary constrained control receives this identical compression; it is not a topology-specific saving.

With token vocabulary size $|\mathcal{A}|$, C allocated contexts, embedding width H, and S slots, the implemented shared model has

$$P_t = 2ND_t + (|\mathcal{A}| + 1 + C)H + S(K_t + 1)(H + 1) + K_t D_t + 4. \tag{18}$$

The final four scalars parameterize the shared geometric deformation. They are absent from the ordinary constrained control. Preallocated context embeddings and every frozen parameter are included. The reported runs use $|\mathcal{A}| = 6$, $C = 6$, $H = 48$, and $S = 2$.

A.2 Learning operations from terminal continuation profiles

For a fixed context, write $F_c(x, w)$ for observed terminal feedback. A history $h = (x, p)$ has a finite continuation profile

$$\sigma_{\mathcal{S}}(h) = \big(F_c(x, ps)\big)_{s \in \mathcal{S}}. \tag{19}$$

The implemented observation table uses the empty word and all one-token words as both prefixes and suffixes. Its successor check also consults words of length three. All such observations come from the supplied TRAIN examples. Different profiles provide explicit state-separation witnesses; equal profiles are a provisional finite-data equivalence, not a claim of equivalence under every future continuation.

The formation routine partitions observed histories by their profiles, checks that observed successors close on the resulting classes, and rejects conflicting or nonreversible successors. If the number of separated classes exceeds D, the workspace grows. An assignment then aligns these classes to the current encoder/readout coordinates. Already learned operations are reused when their actions agree on every constrained reachable state. Otherwise, a constrained matching completes an executable permutation, which is checked against the entire current TRAIN set before neural installation. Candidate search, matching, full-data checks, state expansion, and installation are all part of learning cost. This procedure does not learn from hidden teacher states, but does exploit the deliberately extensive short-word coverage of this benchmark.

After installation, the learner executes real gradient updates of the neural encoder, readout, coordinates, embeddings, and calling network. In the measured streams, new-task validation already equals one immediately after formation and installation, before these updates. Thus the first acquisition in these runs is attributable to terminal-profile identification and installation, rather than to the subsequent 600 Adam updates alone.

A.3 Binding protection and neural–native consistency

Protecting an operation permutation alone does not protect a task: changing the input encoding or call route can destroy an old computation even if every bank operation remains unchanged. The implementation therefore stores admitted native computation references and checks a transition-closed relation between their reachable states and the updated computation. Shared proposals are accepted, backtracked, or rejected according to this binding-level contract. The relation permits state expansion and alternative internal coordinates when they preserve decoded execution. It is a conventional semantic protection condition, distinct from continuous-loop homotopy protection.

The soft interface requires an additional maintenance step. At each task boundary, the implemented consistency refresh sharpens the model's own current encoder, decoder, operation classes, and both original call slots. It obtains no new task labels and adds no parameters. It preserves the current native graph while increasing the probability assigned to its execution path. A sufficient path-mass condition is stated in Section F. Frozen models preserve aliased frozen endpoints during this refresh. The same maintenance is available to ordinary constrained and modular controls.

A.4 Historical six-stage acquisition measurements

Each stream contains six tasks. The first three use unknown reversible N-state operations. The next three require a hidden-memory workspace with $2N$ recurrent states and only N observable output labels. The target hidden mode and transition rules are never provided to the learner. TRAIN contains all initial states paired with every word of length at most three, followed by random words of lengths four through eight: 8,192 examples per task for $N = 8, 16$ and 16,384 for $N = 32$. The larger pool is required to include the $259N = 8,288$ mandatory short examples at $N = 32$.

Validation, IID, OOD, and length-16 evaluations each contain 512 examples per task. IID and validation use lengths at most eight and can overlap the exhaustive short training coverage; reported OOD uses lengths nine through twelve. Length-16 examples are disjoint from TRAIN by length. These are finite held-out computations, not evidence about arbitrary lengths. This historical prototype is bounded to lengths at most 18 and does not supply the required length-32/64 publication measurements.

Define $c_i(t) = 1$ when task i meets both neural and deployed thresholds at the end of stage t: IID accuracy at least 0.95 and both OOD and length-16 accuracy at least 0.90. Admission additionally requires validation

Scale	P_0	P_1	P_2	P_3	P_4	P_5	Completed seeds
$N = 8$	1596	2232	2868	3946	4630	5314	0, 1, 2
$N = 16$	2036	2720	3404	5130	5910	6690	0, 1, 2
$N = 32$	3684	4464	5244	9034	10006	10978	0

Table 6: Historical native/coordinate-renderer streams, retained as internal ablations rather than Primary Prometheus benchmark results. For every completed seed shown, $Q_t = 1, 2, 3, 4, 5, 6$ and every seen-task neural/deployed accuracy equals one on each of the four measured splits at each recorded stage boundary. This table preserves the locally audited historical snapshot; it does not complete the five-final-seed protocol for the future frozen Primary.

accuracy at least 0.95. The strictly retained boundary count is

$$Q_t = \sum_{i \le t} \prod_{u=i}^{t} c_i(u). \tag{20}$$

We report $\Delta P_t / \Delta Q_t$ only when a positive new stable capability is obtained without losing an already stable one. The first acquisition pays the entire initial allocation, $P_{-1} = 0$. This is a stage-boundary criterion, not an unobserved guarantee between measurements. These capability thresholds define a reported outcome, not an accuracy-based eligibility test for baselines.

At $N = 8$, seed 0, the final bank contains 38 learned operations and $D = 16$; the analogous final workspaces are $D = 32$ and 64 at the larger scales. The $N = 32$ runs use a factored realization backend that preserves the learned finite operation semantics without extra trainable parameters. This engineering change and the differing acquisition pools prevent interpreting the table as a controlled timing-scaling law.

Boundary maintenance resolves a measured neural/native mismatch, rather than proving that the neural interface never forgets. In the repaired $N = 8$, seed-0 stream, the minimum recorded old-task soft accuracies in the successive later stages are approximately $0.348, 0.309, 0.252, 0.242, 0.178$, before returning to one at the maintained boundary. Recorded deployed execution remains correct. The original unrefreshed version retained only one task under the strict boundary-stability criterion; its failed trajectory is preserved separately.

The strongest matched finite-program control is decisive for attribution: ordinary constrained compact learning attains the same six-task boundary result at $N = 8$ and $N = 16$, seed 0, with respectively 5,310 and 6,686 parameters, exactly four fewer than the topology model. A repaired frozen-backbone policy and separate task modules also attain six retained tasks at $N = 8$, with 6,384 and 8,664 parameters. Therefore these measurements support compact shared acquisition and a useful consistency repair, but do not isolate a monodromy advantage in parameter growth for this historical version. In particular, 5,314 is not a measured parameter count for the future fully fused Primary. Primary acquisition, retention, parameter growth, and all resource comparisons must be established by actual retraining after the new architecture freeze.

Evidence locations. The frozen seed-0 growth and learning-curve extract is `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/parameter-boundary-growth-evidence.json` and `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/parameter-boundary-learning-curves.csv`. This extract predates the additional seeds. Their original summaries, raw updates, checkpoints, and synchronization hashes are under `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/runs/`, including `parameter-n8-seed1-topology-full-consistent`, `parameter-n8-seed2-topology-full-consistent`, `parameter-n16-seed1-topology-full-consistent`, and `parameter-n16-seed2-topology-full-consistent`. The $N = 32$, seed-0 lineage is preserved in the `parameter-n32-seed0-topology-through0-consistent`, `parameter-n32-seed0-topology-resumed-through2-consistent`, `parameter-n32-seed0-topology-resumed-through3-consistent`, and `parameter-n32-seed0-topology-resumed-consistent` run directories.

B Learning executable monodromy classes

The mechanism studied here separates two operations that are easily conflated: changing the implementation of an existing computation, and acquiring a different executable computation. A deformation that is certified to preserve a based-loop homotopy class can support the former, but cannot by itself accomplish the latter. We therefore develop a learning interface with class-changing proposals for unadmitted operations and class-preserving transactions for admitted operations. The current evidence establishes this interface in a small supplied covering; it does not yet establish an independent topological advantage over equally informed conventional optimization.

B.1 Covering space, marked states, and executable composition

For an integer $D \geq 2$, consider the supplied polynomial family

$$F_D(z,u) = z^D - Dz - u, \qquad \mathcal{C}_D = \{-(D-1)\zeta : \zeta^{D-1} = 1\}, \qquad U_D = \mathbb{C} \setminus \mathcal{C}_D. \tag{21}$$

Projection onto u defines an unbranched D-sheet covering over U_D. We fix the control basepoint $u_* = 0$ and mark its D distinct roots once. Lifting a based loop $\gamma : [0,1] \to U_D$ transports each marked root to another root in the same fiber. Its endpoint action is a permutation $\Pi_\gamma = \rho_D([\gamma])$. For chronological concatenation, defined so that $\gamma_1 \star \gamma_2$ executes γ_1 first,

$$\Pi_{\gamma_1 \star \gamma_2} = \Pi_{\gamma_2} \circ \Pi_{\gamma_1}. \tag{22}$$

An internal state is a sheet index, not a freshly initialized root at every call. Equation (22) consequently specifies an executable, generally noncommutative composition rule.

The executed class-acquisition diagnostic uses $D = 3$. The roots at $u_* = 0$ are labelled $0, 1, 2$ in the order $(-\sqrt{3}, 0, \sqrt{3})$, and the critical controls are -2 and 2. Its one-parameter family is the polygonal loop

$$\gamma_a : 0 \longrightarrow a + i \longrightarrow a - i \longrightarrow 0, \qquad \Pi_{\gamma_a} = \begin{cases} B = (12), & a < -2, \\ I, & -2 < a < 2, \\ A = (01), & a > 2. \end{cases} \tag{23}$$

The singular cases $a = \pm 2$ are not executable. For example, $\Pi_B(\Pi_A(0)) = 2$, whereas $\Pi_A(\Pi_B(0)) = 1$; the order of calls matters. The permutation labels in (23) describe the independently checked outcome, rather than an operation table provided to the learner. The numerical executor actually continues all three roots with a safeguarded predictor/Newton corrector. Its full-fiber endpoint observations, known covering, critical controls, and numerical costs are available to every comparison arm. Thus the protocol combines terminal task labels with full-fiber execution observations; it is not a single-sheet-only observation protocol. The executed source and independent reconstruction are preserved in `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/versions/code-parameter-loop-boundary-learning-v1/loop_boundary_learning.py` and `.deep-work/2026-10-31T234403-0400-v2/supervision/reviews/review-038-loop-boundary-learning.md`.

B.2 What safe implementation deformation does and does not learn

An earlier construction first identifies a permutation, compiles it into a loop of distinct root configurations, and then changes its geometry using a time-dependent ambient shear. Writing a point as $x + iy$, this shear is

$$\Phi_{\omega,t}(x + iy) = x + i\left[y + \sin^2(\pi t/T) h_\omega(x)\right], \qquad h_\omega(x) = \sum_\ell \omega_\ell \tanh(v_\ell x + b_\ell), \tag{24}$$

where v_ℓ, b_ℓ are fixed. Every finite ω gives a global diffeomorphism, with an explicit inverse obtained by subtracting the same vertical displacement. At both execution endpoints it is the identity. Interpolating ω therefore gives a collision-free homotopy of *configuration-space* loops with a fixed base configuration, preserving their braid and endpoint permutation. This is a useful within-class learning freedom, but it is not

a mechanism for discovering a new permutation. It is also a different construction from the control-plane covering in (21); the two spaces must not be identified without an additional map between them.

The new class-learning interface instead trains the raw vertices of a loop in the fixed control plane. A controller $a_\theta(e)$ maps a public operation feature e to coordinates in a supplied affine polygon family. For an unadmitted operation, a finite update $\theta \mapsto \theta^+$ may place the two regular endpoint snapshots in different chambers. The parameter interpolation between them may pass through a singular loop, but that interpolation is not executed. Only the separately checked regular snapshots are run. There is consequently no claim that a nonsingular based homotopy can change monodromy. The covering, basepoint, critical set and sheet labels remain fixed. Moreover, a change of homotopy class in the kernel of ρ_D would not count as a new executable operation.

B.3 Terminal boundary feedback and neural credit

An exact endpoint-semantic loss is locally constant in regular loop coordinates. Ordinary within-chamber differentiation alone therefore provides no direct derivative for acquiring a different endpoint action. Let V be raw loop coordinates drawn from a smooth location kernel centered at a_θ, and let $\ell(V;\theta)$ denote the terminal loss obtained by executing that loop and the current neural bindings. Define

$$J_\sigma(\theta) = \int \ell(v;\theta) p_\sigma(v - a_\theta)\, dv. \tag{25}$$

For the current fixed-cover, endpoint-semantic objective, the coupled derivative has two contributions:

$$\nabla_\theta J_\sigma = \mathbb{E}[\partial_\theta \ell(V;\theta)|_{V \text{ fixed}}] + (D_\theta a_\theta)^\top g_{\text{boundary}}. \tag{26}$$

The first term trains the input encoding, readout, calls and their shared backbone while holding the actually observed endpoint maps fixed. For a coordinate slice with crossing events b_r and terminal-loss jumps $\Delta\ell_r = \ell(b_r^+;\theta) - \ell(b_r^-;\theta)$, the second term uses

$$(g_{\text{boundary}})_j = \mathbb{E}_{V_{-j}}\left[\sum_r \Delta\ell_r\, p_\sigma(b_r - a_{\theta,j})\right]. \tag{27}$$

Safe representatives of the neighboring chambers are actually executed to obtain these terminal losses. The implementation uses a normalized raised-cosine kernel and charges event detection, representatives, terminal callbacks and continuation. A coordinate moving one free vertex affects its two adjacent segments; tied coordinates moving several vertices are charged for every affected segment instead of silently inheriting the smaller bound.

Equations (26)–(27) are applications of known differentiation of piecewise-smooth objectives, not a claimed new calculus theorem. A path-work objective with smooth within-chamber dependence on v would require its corresponding continuous derivative as well. The implemented boundary term changes how probability mass is assigned to executable classes; it does not replace successful discrete execution by a mixture of several operations.

The ideal smoothing distribution assigns mass arbitrarily close to a discriminant. The numerical executor additionally requires positive finite clearance. These are distinct domains. The initial score-function arm was censored by a probe entering the exclusion band, rather than assigned a fabricated failure loss. The later hybrid interface evaluates an ideal sample's class at a paid safe representative when available, while still requiring the deterministic candidate loop itself to be executable. This scalar-family convention is exposed to the conventional controls and is not assumed to solve the general high-dimensional feasibility problem.

B.4 A sufficient protected transaction

For an admitted polygonal loop γ, let $\delta = \min_{t,c\in\mathcal{C}_D} |\gamma(t) - c| > 0$. If its updated vertices satisfy

$$\max_k |v_k(\theta^+, e) - v_k(\theta, e)| < \delta, \tag{28}$$

with the same segmentation and fixed endpoints, linear interpolation gives a based homotopy avoiding $\mathcal{C}_D$. Indeed, the displacement of every point on every interpolated segment is bounded by the maximum vertex

displacement; its distance from the critical set remains positive. Hence its endpoint permutation is unchanged. Preserving each admitted generator then preserves every word of those generators under fixed input/output bindings.

This sufficient topological certificate is accompanied by separate checks of the executor's coordinate box and numerical clearance. In the diagnostic the minimum allowed control clearance is .015. Protection of a permutation alone does not protect a changing encoder, router or decoder; the additional binding and soft-execution contracts are specified in Section C. None of these conditions implies hardware noise protection or a quantum energy advantage from a CPU simulation.

B.5 Executed class acquisition and the unresolved advantage

The initial scalar diagnostic uses two real shared weights,

$$a_{\text{old}} = w_0, \qquad a_{\text{new}} = w_0 + w_1, \qquad (w_0, w_1) = (-3, 4). \tag{29}$$

The old operation B is supplied as the starting mechanism and verified by continuation; it is not credited as learned in an earlier stage. Four terminal training cases require A, which is outside the admitted subgroup $\{I, B\}$. Two accepted neural updates produce

snapshot	a_{old}	a_{new}	training error	old error
initial	-3	1	.75	0
first	-2.625	1.75	.75	0
second	-2.26427259	2.47145482	0	0

The old loop actually changes geometry while retaining B; the new loop crosses its class boundary and acquires A. Four frozen held-out composition cases and an actual checkpoint reload also have zero error. These are finite length-at-most-three tests, not a scaling or long-term continual-learning result. The raw execution is `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/runs/parameter-loop-boundary-learning-v1/learn/result.json`.

The equally informed ordinary event optimizer produces the same weights, parameter hashes and counted trajectory. Discrete chamber search succeeds with one update, 35 program executions and 37 terminal callbacks, versus 70 executions and 72 callbacks for the event arm. Those totals include their disclosed calibration and audit work. A reexecution guard spends more continuation work than the analytic certificate on this instance, but the ordinary optimizer is also allowed that certificate. Thus the result closes the earlier map-first learning gap, while the principal question remains open: does monodromy organization improve acquisition, retention, verification or parameter growth beyond the strongest ordinary constraint and search methods under matched information and complete cost?

C A trainable loop-based Prometheus Block

The purpose of a Prometheus Block is to join operation formation with executable state updates, learned calls, composition and retention in one trainable state. The present implementation is a concrete fixed-three-state prototype of that interface. It removes the requirement to learn a permutation matrix first and render a loop afterward, but has not yet become a general computational architecture. We specify its actual computation and its observed failure before describing the remaining interface requirements.

C.1 Neural state and computation

The current `HybridLoopBlock` stores a trainable input encoder, a trainable readout, call heads and a shared loop controller. A public feature e is transformed by a shared matrix H; a loop head r produces

$$h_\theta(e) = He, \qquad a_\theta(e) = r^\top h_\theta(e), \qquad \gamma_{a_\theta(e)} : 0 \to a_\theta(e) + i \to a_\theta(e) - i \to 0. \tag{30}$$

The two operation features and two token features are the disclosed vectors $(1, 0)$ and $(1, 1)$. The same H participates in both loop generation and calling, so changing a new operation can change the old operation's

geometry and the call probabilities. For call slot s and input token b,

$$p_{\theta,s}(k \mid b) = \mathrm{softmax}(C_s h_\theta(e_b))_k, \qquad k \in \{\mathrm{skip}, 0, 1\}. \tag{31}$$

Skip supplies an identity computation. The other two choices refer to loop bank entries. There is no trainable permutation-score tensor W_k in this prototype: endpoint maps are derived from paid actual root continuation.

Let E_θ and R_θ be row-stochastic matrices obtained from the encoder and readout logits. For an input x and a token word $b_1 \cdots b_T$, the soft learning path is

$$q_0 = E_\theta[x, :], \tag{32}$$

$$q_{t,s+1} = q_{t,s}\left[\sum_k p_{\theta,s}(k \mid b_t) M_k(V)\right], \qquad M_{\mathrm{skip}} = I, \tag{33}$$

$$p_\theta(y \mid x, b_{1:T}, V) = [q_{T,S} R_\theta]_y, \tag{34}$$

where $M_k(V)$ is the one-hot matrix of the actually observed endpoint permutation for the sampled loop coordinates. These matrices are held constant during the within-chamber derivative in Equation (26). The terminal training loss is $-\log p_\theta(y)$, with a declared numerical probability floor of 10^{-12}. The target supplies only the final task output, not a correct route or an intermediate state.

Native execution uses the encoder, call and readout argmax choices and composes the resulting permutations on the same sheet. This is the deployed discrete program against which task success is checked. Cached maps are derived observations of actual loops, not separately learned operations. Reusing a receipt implements the same endpoint semantics without claiming that the changed continuous trajectory has been freshly integrated. Cache namespace, payload, root observations and cold-reexecution audits are recorded and made available to conventional controls.

The executed one-slot model has 30 trainable scalars: 9 encoder logits, 9 readout logits, 4 shared-backbone weights, 2 loop-head weights and 6 call-head weights. It also stores 8 fixed feature-buffer scalars. Both the old operation and its call are supplied initial conditions, while the new call begins uniformly across all three choices. Encoder/readout logits begin near the identity but remain trainable. This starting ability is disclosed rather than counted as previously acquired learning. The exact executed source is `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/versions/code-parameter-hybrid-loop-learning-v1/hybrid_loop_block.py`.

C.2 One learning transaction, three retention obligations

The persistent state comprises neural parameters θ, admitted loop receipts, a reference native binding relation, bounded derived-map storage and the numerical execution contract. Given current terminal examples, a learning transaction proceeds as follows.

1. Sample the unadmitted operation's loop coordinates. Execute or reuse legitimately certified maps, charge their observations, and evaluate the actual soft terminal computation.

2. Compute within-chamber gradients for encoding, readout and calls, together with the class-boundary term. Both derivatives pass through the same neural backbone; no correct new call is inserted manually.

3. Propose an actual shared-weight update. Backtrack its finite scale while checking numerical executability, admitted loop protection, old native binding closure and soft/native agreement. Rejected executions and checks remain in the cost ledger. An aborted transaction restores the original parameter values in place.

4. Admit a new operation and its token binding only after deterministic mean-loop execution gives perfect native and soft accuracy on the stated training admission set and satisfies the expanded confidence contract. The reference then records the actually acquired program, rather than a teacher operation table. Generalization remains a separate held-out test.

These checks discharge different obligations. First, geometric clearance protects the endpoint action of an old loop, as in Equation (28). Second, an old task may still be forgotten if its encoder or call changes,

even when every bank permutation is unchanged. The implementation therefore computes a finite relation $\mathcal{R}$ between the reference and proposed native states, starting from all old encoded input pairs, and requires

$$\begin{aligned} &(E^-(x), E^+(x)) \in \mathcal{R}, \\ &(s,t) \in \mathcal{R} \implies R^-(s) = R^+(t), \quad (T_b^-(s), T_b^+(t)) \in \mathcal{R} \quad (b \in \mathcal{A}_{\text{old}}). \end{aligned} \tag{35}$$

Here $E^{\pm}, R^{\pm}, T_b^{\pm}$ denote the complete hard native encoder, readout and token transition before and after the update, including all call slots. Exhausting this finite product-state closure preserves the decoded behavior of every finite old word. It does not require the two internal representations to use identical state labels. This is a conventional finite behavioral-relation check; its correctness is not a topology-exclusive claim.

Third, native preservation alone is insufficient for the soft neural path. Let $\underline{p}_E$, $\underline{p}_R$ and $\underline{p}_C$ be the minimum probabilities assigned to the model's own hard encoder, readout and old-token call choices. Because every executed bank map here is exactly one-hot, the probability of following the entire own-native path through a word of length at most L is bounded below by

$$m_L = \underline{p}_E \, \underline{p}_R \, \underline{p}_C^{SL}. \tag{36}$$

If $m_L > 1/2$, the soft output has the same unique argmax as that native program: the designated path alone carries more than half of the total probability. The current transaction uses a small additional tolerance above $1/2$ and $L = 16$. The mathematical argument is real-arithmetic; the reported float64 evaluation is not interval certification. Native closure covers all finite old words, whereas this particular soft bound covers the specified finite length. Keeping these scopes distinct prevents a finite confidence check from becoming an unsupported unlimited-length neural guarantee.

C.3 Composition and a constructive module interface

The executable interface is a state transformation plus a receipt:

$$(s, b, \theta, \mathcal{M}, \mathcal{K}) \longmapsto (s', y, \text{receipt}), \qquad s' = \Pi_{k_S} \circ \cdots \circ \Pi_{k_1}(s), \tag{37}$$

where $\mathcal{M}$ is the admitted mechanism state and $\mathcal{K}$ contains the execution and protection contracts. Consecutive calls reuse the output sheet, making AB, BA and longer words genuine computations rather than independent classifications. Learning can revise the shared controller and the call structure; admission converts a successfully acquired computation into a protected reusable entry. Stored receipts permit certified semantic reuse, with their maintenance and storage charged separately from parameters.

This interface also identifies the obligations for a general Block. A layer must expose which state representation it passes to the next layer, and learning must preserve the composed input–operation–readout binding, not just each isolated loop. A capacity change must provide a state embedding, transferred operation semantics and an updated protection relation. An irreversible operation would require a disclosed reversible extension or a different state update, charging the extra workspace, cleanup and readout. Continuous inputs require a learned encoding and an explicit domain on which its binding is protected. These are implementable interface requirements; they are not achieved merely by calling the present $D = 3$ module a general Block. The prototype currently supplies fixed finite input features and a fixed covering, and has not implemented capacity growth, noninvertible computation, continuous-domain generalization or verified multilayer stacking.

C.4 Executed hybrid V1: operation formation without task acquisition

Hybrid V1 is an actual negative architecture-development result. The frozen experiment uses four terminal training cases, four held-out composition cases, a supplied old operation, and 100 parameter transactions per arm. The new call is initially uniform; success cannot be obtained by receiving a correct route. The three arms apply the coupled gradient, only its within-chamber component, or only its boundary component. Both components are nevertheless computed and charged in all arms, so their run times are not an ablation efficiency comparison.

arm	updates	no-op	map	TRAIN (S/N)	held-out (S/N)
coupled	100	72	A	.25/.25	0/0
within only	100	0	I	.25/.25	0/0
boundary only	100	94	A	.25/.25	0/0

All arms retain old-task soft and native accuracy 1, and none admits the new operation–binding pair. Acquiring the endpoint map A therefore does not establish acquisition of the task that should call it. A "transaction" is explicitly a committed proposal, including a zero-scale proposal; it must not be reported as 100 nonzero learning updates. Raw curves and actual neural checkpoint reloads are preserved in `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/runs/parameter-hybrid-loop-learning-v1/learn/result.json`.

The coupled arm exposes a specific repair target. At its final snapshot the within-chamber call-head gradient norm is approximately 1.2583, but every tested positive joint step scale, down to 1/64, is rejected by the old-loop geometry/map guard; the transaction commits scale zero. Thus learning credit exists while this single-direction transaction cannot use it. The observation does not prove that every feasible update is blocked. It motivates testing a component-wise or constrained repair that can preserve an admitted class without discarding usable binding credit. Such a fallback is development work, not an executed V1 result, and no outcome is predicted here.

The implementation accounts for terminal callbacks, root continuation, event geometry, neural vector–Jacobian products, proposals and rejections, finite relation checks, cache bytes and real checkpoint storage. Ordinary controls receive the same mathematical certificates and execution observations. Consequently neither the small parameter count nor old-task retention in this failed acquisition test is evidence of an independent Monodromy advantage. The concrete contribution at this stage is the unified trainable interface and an observed, localized failure of its protection– plasticity transaction, with the general-Block and competitive-benefit questions still open.

C.5 Executed hybrid V2: a targeted feasible-direction repair

V2 retains V1's loss, examples, thirty trainable scalars, eight buffer scalars, and protection certificates. Its only transaction change is to try the same gradient on the existing encoder, readout, and call heads when all positive joint backtracking scales are rejected. The backbone and loop head stay fixed in that fallback; every proposed update still passes all three certificates. This is a disclosed block-coordinate repair, applicable to ordinary constrained optimization as well as this topological realization.

The complete arm finishes after 31 gradient transactions and one own-class confidence-maintenance transaction. Three commits use the head-only fallback. TRAIN and the four held-out composition cases attain soft/native accuracy 1; old observed cases remain at 1. Both single-component arms exhaust 100 transactions without acquiring the new calling behavior. The first 28 complete-arm transactions reproduce V1 exactly. New class acquisition occurs before the eventual calling success, distinguishing the two learning events.

The complete arm uses 122 executed programs, 716 terminal callbacks, 11,526 continuation steps, and 127 proposals, of which 95 are rejected. The one maintenance step sharpens its own acquired hard route without adding parameters or receiving a teacher route. Its length-16 designated-path mass rises from approximately 1.53×10^{-5} to .999779. This validates the specified interface on the diagnostic, not an architecture-level advantage or a general multi-stage learner. Both gradient components are computed and charged in every arm; their measured times do not rank optimized ablation efficiency.

The complete neural checkpoint and cold operation maps were actually reloaded. Independent review039 checked all 67 synchronized source/result files, model tensors, raw curves, and the common prefix. The frozen run is `probes/topology-architecture/runs/parameter-hybrid-loop-learning-v2/` within the V2 campaign. This development component is not a frozen Primary Prometheus and supplies no primary five-layer benchmark numbers.

C.6 Fourth-phase two-address development (executed, not Primary)

The earlier held-out query skeleton can overwrite its first read. A new, prospectively frozen development phase therefore uses two independently written values, independently varying write and query orders, and terminal feedback in which each stored value causally changes the output. Correct operation semantics, routes, intermediate states and counterfactual truth are excluded from learner inputs. TRAIN/VALID contain 4096/512 examples; IID, long (64–112 tokens) and held-out distractor tests contain 512 each. The balanced $8 \times 8 \times 2 \times 2$ design puts TRAIN-majority and last-four-token lookup controls at 12.5%.

Each arm resumes its own actual three-phase NN, Adam, RNG and 128-row replay state. A single randomly initialized token embedding row is added: 16 parameters for each continuous Block, 44 for GRU and 32 for Transformer. Old embedding rows, optimizer moments/steps and old-token forward values are checked exactly before learning. All four actual CPU growth/reload checks passed. The only new phase uses 600 updates of 48 current plus 16 replayed examples; prior training is not rerun.

The dynamic Block's actual completed phase has 28,381 parameters, VALID soft/hard-call accuracy of 58.20/33.20%, and IID soft accuracy of 56.64%. Its previous-task IID accuracies fall to 79.30/82.03/78.91%. Actual training wall time is 91.42 seconds for 38,400 examples (28,800 current, 9,600 old); training-forward work is 29.446 billion MAC proxies, which excludes backward FLOPs. Recorded TRAIN/VALID curve evaluation takes another 42.70 seconds; test, replay, restore, optimizer, checkpoints and inference are separately charged in the raw ledger. Total backward FLOPs and energy are not inferred from MAC proxies. Other arms are separately executed under the frozen protocol; this interim result is not a competitive advantage or a frozen Primary claim.

`memory_live_stage_train.py` and its frozen protocol, raw updates and actual checkpoints are retained in the same V2 campaign.

D TODO 4: Transfer to Real-World Continual Learning

The same core computation and learning rule must be evaluated on algorithmic, audio, and visual tasks. Input adapters and output heads may change, but changes to the core block or its protection rule define an architecture variant and must be reported separately. Synthetic permutation accuracy alone does not establish this transfer.

D.1 Fixed audio protocol

Speech Commands v2 uses all 35 word classes in seven class-incremental stages, with five new classes at each stage. Training, validation, and test examples follow the official split and all methods use the same 16 kHz input preprocessing. Each stage trains only on its permitted current data and declared memory. Evaluation recognizes all classes seen so far without a task or class-group identifier at inference.

D.2 Fixed visual protocol

Split CIFAR-100 uses ten class-incremental stages, each introducing ten fine-grained classes. The official 50,000-example training set supplies a per-class 10% validation subset under a fixed seed; the official 10,000-example test set is not used for development. All methods share preprocessing and augmentation rules and use a single expanding classifier without a task identifier at inference.

D.3 Memory, initialization, and cost

Both tasks require Memory-0 and Memory-500 comparisons. The latter permits at most 500 old examples across all classes. Saved features, labels, teacher outputs, statistics, and other persistent information are disclosed and charged separately; a representation cache is not free memory. Main experiments use random initialization. Pretrained runs, if any, form a separate comparison with matched pretraining access. The fixed per-stage epoch budgets are 10, 50, and 200, with validation-only configuration selection under the current benchmarking protocol.

D.4 Evidence boundary

No audio or visual training result is available in the present checkpoint. In particular, the existing synthetic learner's public task embeddings cannot be passed to these class-incremental evaluations as inference-time task identifiers. The transfer study must test whether the learned computation actually reduces acquisition, retention, parameter, or full lifecycle costs under these protocols.

Layer	Required scientific coverage	Current evidence boundary
I	Six-stage acquisition at $N = 8, 16, 32$	Historical streams in Table 6; the frozen fully fused Primary must be retrained under the five-seed protocol.
II	Noncommuting programs, unseen combinations, lengths 16/32/64, dynamic write/overwrite/read/delay memory	Finite held-out words and length 16 tested; the complete complex-program and dynamic-memory suite is not completed.
III	One Block for reversible, nonreversible, continuous, and variable-length computation	Current finite reversible operations do not establish this generality.
IV	Speech Commands v2 and Split CIFAR-100	No competitive real-data result is claimed in this manuscript checkpoint.
V	Applicable architectures, continual-learning controls, and mechanism ablations	Historical valid measured configurations and working program controls; prospective bounded family comparisons remain incomplete.

Table 7: All five layers require direct measurements. Evidence at one layer does not replace another layer.

E Benchmarking and competitive evaluation

Primary architecture and historical evidence. Primary Prometheus is the latest fully fused architecture actually validated under TODO 3. TODO 2 does not block the Primary freeze; a component without measured benefit may be omitted. Its core mechanism, code version, and evaluation configuration are frozen before actual retraining across all five benchmark layers. The earlier native/coordinate-renderer streams are historical results and internal ablations only; their outcomes, parameters, and costs are not pooled with or substituted for new Primary measurements. The limited D3 hybrid V2 diagnostic is development evidence, not completed TODO 3 validation or a Primary freeze.

Baseline validity is independent of accuracy. A configuration is valid evidence when its implementation is checked, any claimed reproduction is faithful, information is fair, and its prospective training/tuning protocol is followed and disclosed. No accuracy threshold determines validity: 100%, 37%, and random-level accuracy are all retained outcomes of valid configurations. Low performance can support a conclusion conditional on the tested model, information, and budget; it does not establish that a complete algorithm family has failed. An equal-quality parameter comparison is simply unavailable when the measured qualities differ, rather than making either run invalid. All failures, configuration trials, retries, maintenance, and extra training remain charged. A low result is not a software defect and does not trigger trials until a desired number appears.

E.1 Five-layer evaluation scope

Layer II will distinguish operation-order dependence, including $AB \neq BA$ and commutators, from memorization of an observed word distribution. Programs are trained at length at most eight and evaluated separately at lengths 16, 32, and 64. A separate key–value stream requires writing, overwriting, reading, and delayed queries. Every architecture receives terminal task feedback without teacher intermediate execution traces. Layer III keeps the Block's core state-transition and learning mechanism fixed across task types; a change to that core is reported as an architecture variant, rather than silently counted as generalization.

E.2 Information, development budgets, and final evaluation

The synthetic comparison uses the same terminal data, stage order, public context, and evaluation protocol. Dedicated constructors, teacher operator tables, hidden state labels, and correct routes are not free priors for any method. Formation passes and terminal-label accesses beyond minibatch training are charged explicitly. Equal gradient-update counts do not imply equal total learning compute when one method additionally solves a full observation table.

The locked synthetic update limits are B1: 600, B2: 2,400, and B3: 9,600 gradient updates per stage, available to every method including Prometheus. Each main model family has at most 24 development configurations: development seed 0 selects candidates, the top three are checked with development seed 1, and the ranking uses a frozen validation criterion. After configuration lock, the final evaluation uses seeds $0, 1, 2, 3, 4$ without test-based tuning. Shared early checkpoints and resumed run fragments count as one lineage, not multiple independent seeds. The real-data budgets are 10, 50, and 200 epochs per phase under the same validation-only selection principle.

Only a localized, reproducible software, data, or training-logic error qualifies as a bug fix. Optimizer, architecture, curriculum, or initialization changes are configuration development unless they repair such a demonstrated error. Any supplemental trajectory outside the registered main budget is disclosed separately. Reaching the highest budget without acquisition remains an allowed negative result. Test and OOD measurements do not select configurations, training duration, or stopping points.

Three comparisons are kept distinct: identical information/protocol, matched training and total-compute budgets, and equal attained capability quality. The final analysis requires accuracy–parameter, accuracy–training-cost, and accuracy–inference-cost frontiers. Different method families may use the optimizers and schedules appropriate to their original algorithms; forcing one recipe on all competitors is not a fairness requirement.

E.3 Fixed real-data protocols and strong competitors

Speech Commands v2 uses 35 word classes in seven class-incremental phases of five classes. It uses the official train/validation/test split and common 16-kHz preprocessing. Each model classifies all seen words at inference without a phase identifier or true class group. Split CIFAR-100 uses ten phases of ten fine classes, the official 50,000 training and 10,000 test images, and a fixed class-stratified 10% training holdout for validation. It uses an expanding single head and supplies no task ID at inference. Input processing, augmentation, and data access are matched across architectures.

Both datasets have Memory-0 and Memory-500 conditions, where 500 is the total number of stored old examples across all classes. Stored features, teacher outputs, task statistics, and other historical state are additionally recorded; they are not an exemption from the memory comparison. Main runs start from random initialization. Pretrained variants are supplemental and give every method the same pretraining information. These no-ID class-incremental protocols are distinct from the public-context synthetic benchmark.

The primary competing architecture family is the Transformer: vanilla, replay, adapters, and replay plus adapters, at two or more prospectively selected scales. Parameter-matched and quality-matched comparisons are both required. Continual-learning comparisons include DEN, PackNet, replay, and ER-ACE or an equally mature class-incremental method. PackNet task-incremental results, when applicable, are explicitly separated from no-ID class-incremental rankings. Mamba-2 participates in appropriate sequence/memory tasks, and Titans in tasks suited to its memory interface. HOPE/Nested Learning participates only with a verifiable, applicable implementation. An unfaithful reproduction or untested implementation is marked `NOT_VALIDATED`, with its specific fidelity or implementation gap stated. Low accuracy alone never assigns that status.

E.4 Measured baseline results at the historical B1 budget

All four runs in Table 8 have zero tasks passing the dual-interface capability criterion. Their low first-task performance shows that the result is not explained by forgetting alone. The $N = 16$ width-64 Transformer similarly has no passing task in the measured stream. These are valid budget- and configuration-conditional observations. They do not by themselves characterize every Transformer configuration or establish an equal-quality parameter ratio, and they are not new Primary Prometheus results.

A separately recorded anchored Transformer trial uses a 100-update source prefix and 2,200 additional updates, totaling 2,300 updates. It is a supplemental configuration trajectory, not a software bug fix or an equal-600-budget result. It is valid configuration evidence under its own disclosed supplemental protocol. The fixed development progression stops at its prospective budget. The validation-selected frozen checkpoint is then tested once, obtaining IID 0.37109375, OOD 0.134765625, and length-16 0.111328125. This improves some short-word behavior without establishing full-task competence; the low test result is not a trigger for unbounded additional trials.

Method	Total P	First task			New task at phase 6		
		IID	OOD	L16	IID	OOD	L16
Transformer-64	103752	.1836	.1230	.1348	.1152	.1094	.1016
Transformer-64+Replay	103752	.1836	.1230	.1348	.1191	.1172	.1055
Transformer-96	229352	.1348	.1094	.1035	.1328	.1406	.1191
Transformer-96+Replay	229352	.1348	.1094	.1035	.1133	.1289	.1152

Table 8: Actual $N = 8$, seed-0 historical streams, each with 600 updates per phase and 3,600 updates overall. "Phase 6" is the newly introduced task at the final phase, not an average over old tasks. These are valid measured B1 configurations under their reported historical protocol, regardless of their low acquisition. They are not pooled with a future frozen Primary evaluation.

Policy	Weight bytes	Extra-state JSON bytes	Optimizer tensor bytes	Checkpoint bytes	Old guard seconds
Topology compact	21256	208326	42544	341373	47.4276
Ordinary constraints	21240	104105	42508	202737	45.6787
Frozen + readout	25536	83598	7568	146821	0
Private modules	34656	81825	12316	160265	0

Table 9: Historical final-phase $N = 8$, seed-0 resource fields for the native/coordinate-renderer policies, not the future fully fused Primary. The guard covers the phase's 600 updates. These are format-specific fields and overlapping cost scopes, not a disjoint sum or a controlled wall-time ranking.

The first 600-update task runs of the adapted DEN-RNN and linear self-modifying CMS reconstruction obtain IID 0.115234375 and 0.138671875, respectively. Their parameter counts are 1,800 and 81,224. The former adapts DEN-style selection, expansion, splitting, and timestamps to a recurrent terminal-feedback interface; the latter is not an official HOPE reproduction. These are valid executed adaptations under their reported 600-update protocols; their low scores do not invalidate them. Conclusions apply to these particular DEN-RNN and CMS variants. Faithful full-original DEN and HOPE reproduction remains `NOT_VALIDATED` because that fidelity has not been established, not because an accuracy threshold was missed.

For the historical finite-program question, matched conventional controls are available. At $N = 8$, seed 0, ordinary constrained compact learning matches all six stage-boundary capabilities using 5,310 parameters, versus 5,314 for the topology model. Repaired freezing uses 6,384 and separate task modules 8,664. The frozen repair adds a compatible 128-parameter readout when terminal-profile output multiplicities require it, and later tasks reuse that readout. Its initial failed version remains recorded. These controls explain the historical shared/compressed parameter benefit without attributing it to monodromy; they complement rather than replace the required Transformer comparison.

E.5 Costs and causal attribution

All stored and frozen parameters count. Reference arrays are contained within extra-state metadata; cache maps and certificates can also share storage. Checkpoint sizes include optimizer, RNG, and container state. JSON size is not minimal sufficient information or peak RAM. Phase wall time includes nested training, evaluation, and protection scopes, which are not added twice. Some jobs overlap on the GPU, so their individual wall times do not support a throughput ranking. One ordinary-control initial-stage report is missing post-training timing fields after a reload-helper failure; these remain null, rather than being reconstructed as measured values.

Complete comparison additionally requires development and failed-run costs, primitive discovery, data accesses, solver and installation work, temporary dense tensors, replay, cache preparation, old-binding checks, rollback, and checkpoint reload. Controlled same-hardware training FLOPs/GPU time, end-to-end inference p50/p95, throughput, and peak memory are not yet complete. Neither software simulation nor these parameter counts measure hardware energy or establish a quantum advantage.

The causal matrix removes, separately, primitive reuse, sharing, workspace growth, old capability protection,

graph constraints, and native compilation. Topology-specific comparisons substitute ordinary constraints for monodromy, disable cross-class loop learning, disable within-class safe deformations, and compare ordinary cross-class optimization and discrete search under the same information and costs. Shared compression, program formation, graph closure, and generic consistency repair are not credited to monodromy. A same-mechanism reimplementation tests executable correspondence; it is not by itself a historical-priority finding or a general impossibility result.

E.6 Statistical reporting and completion status

Formal comparisons use all five independent final seeds, mean, standard deviation, and 95% confidence intervals; paired-seed differences and their intervals are primary. Multiple primary tests receive multiplicity correction, and failed seeds remain included with their actual censoring or failure status. The principal outcomes are final/average-incremental accuracy, forgetting, backward transfer, acquisition cost, and total parameters and resources at equal retained quality. The earlier scale/seed lineages remain historical or internal ablations; they cannot count as retraining of the newly frozen Primary, be pooled across architecture versions, or supply its five-seed statistics.

Publication-grade benchmarking remains `INCOMPLETE`: the fully fused TODO 3 must pass; the complete architecture is then frozen and actually retrained across every layer against protocol-valid competitors under the bounded development procedure. The two fixed real-data benchmarks, causal attribution, and full resource/statistical evidence remain required. No competitor must cross an accuracy gate to have its measured result included, and no favorable historical finite-task result substitutes for the new Primary evaluation.

Evidence locations. Original neural streams and their actual checkpoints reside under `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/runs/`, including `parameter-n8-seed0-transformer64-full`, `parameter-n8-seed0-transformer64-anchored-competence-v1`, `parameter-n8-seed0-den32-through0-v2`, and `parameter-n8-seed0-selfmod_cms64-l1-through0-v2`. The frozen stage-level extract is `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/parameter-boundary-growth-evidence.csv`. Actual original-source reading and adaptation boundaries for DEN are in `.deep-work/2026-10-31T234403-0400-v2/sources/parameter-den/reading-and-adaptation.md`. The governing prospective benchmark protocol is `ALL.md`, TODO 5, with the latest explicit human directives controlling architecture identity and accuracy-independent configuration validity if older wording conflicts.

E.7 Observed continuous-core development: parameters and full cost

All new development reports jointly expose parameters and relevant computational cost. Forward MAC proxies exclude backward, optimizer, indexing and normalization; they are not total training FLOPs. Energy and end-to-end serving latency are unmeasured, not zero. Nested timing fields are not added twice.

All-token model	P	PPL64	Train s	Driver s	Forward G MAC
Dynamic	41,593	10.358	75.96	119.47	21.715
Static	41,593	10.306	75.24	118.74	21.715
Detached	41,593	10.243	74.07	114.41	21.715
Transformer36	39,664	10.533	6.75	12.26	29.816
GRU44	32,660	8.664	2.49	7.96	16.977

These actual runs use300 updates of32 length64 windows:614,400 next-character label observations per method. This is a separate protocol from the earlier 19,200 terminal-prefix labels. All-token and common-suffix NLL vectors, input hashes, actual checkpoints, Adam state and curves are preserved. Driver time includes measured validation, test and audit work but is not a complete machine startup time; external runner receipts include import/CUDA overhead. Training wall times were collected under the recorded possible concurrency and are not exclusive-throughput rankings. Optimizer bytes are332,844 for each Block, 317,476 for Transformer, and261,324 for GRU; allocated buffers, input cache, checkpoint files and allocator peaks are separately reported in `alltoken-parameter-complete-cost-evidence-v1.json` within the campaign probe directory. The GRU result is a strong concrete alternative, not an accuracy-gated baseline.

The64-dimensional real-input interface adds10,560P on speech and10,944P on images, to42,840/51,737P. Actual dynamic TEST accuracy is72.27%/42.20%, versus the previous16-dimensional29.77%/42.10%; static speech already achieved71.17% with32,280P. Thus additional capacity and gradient dynamics must be distinguished, and none of these changes demonstrates a broad resource win. All six interface-width runs, all earlier failures and strong controls remain separate records, not selected best-of-run Primary statistics.

The context Replay128 intervention keeps600 updates and64 examples per batch, using48 current and16 stored old examples. The total128-row raw/provenance buffer costs55,296B in each arm; duplicates consume capacity and are disclosed. Actual old-task0 final IID rises from82.81% to94.14% for the dynamic Block, versus96.68% for GRU with the same buffer. Formation/search is absent from this fixed-bank variant; this fact does not erase the historical discovery cost of other variants or the costs of development and failed executions.

Memory dependence and measurement scope. A frozen28,365P context checkpoint underwent36 VALID-only interventions, with9,216 label evaluations and zero learning. Third-stage soft/balanced accuracy was95.31%/90.28% intact,82.81%/62.63% with zero readout, and87.89%/74.03% after value-only slot rotation. Joint key/value rotation preserved all predictions, with floating-point probability differences retained. All extra read/instrumentation work and the first failed logging attempt are charged. This establishes frozen execution dependence on learned key–value pairing, not improved acquisition or retention. The earlier unseen skeletons can erase the effect of an intermediate read; they do not independently establish two-address composition. A new prospective live-read protocol is being constructed without altering the earlier data or results.

Figure provenance. The following plots are generated from fixed JSON/JSONL bytes and retain their original protocols. In particular, the character figure is the earlier terminal-prefix experiment, not the new all-token table. Source bytes, plotted arrays/CSV, failed-run inventory and hashes accompany `parameter-development-figures-v1`. No plot carries across-seed confidence bands because these are single development trajectories.

E.8 Fourth-phase and same-weight execution cost figures

These figures are descriptive single-seed development evidence. Parameters, training-forward MAC proxies, observed timings, optimizer/replay/buffer payloads, checkpoint storage and allocator peaks are shown jointly. Different scope timers are not additive. The character-LM execution audits keep weights fixed and use no labels; they are separate from the fourth-phase learning task. The data/source hashes and all failed attempts remain archived.

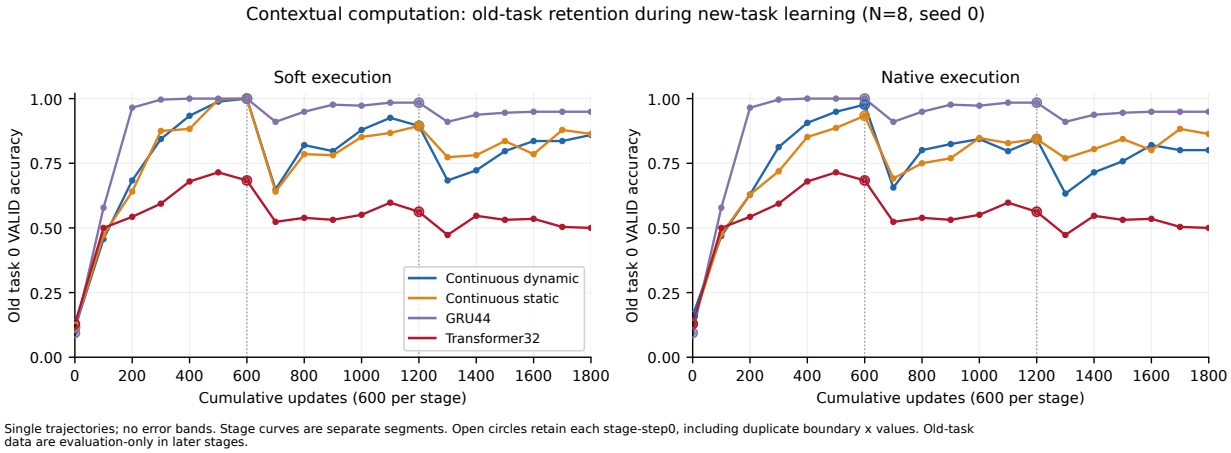


Figure 1: Actual contextual old-task VALID curves, without replay. Open stage boundary points and transient forgetting are retained. This is not Primary.

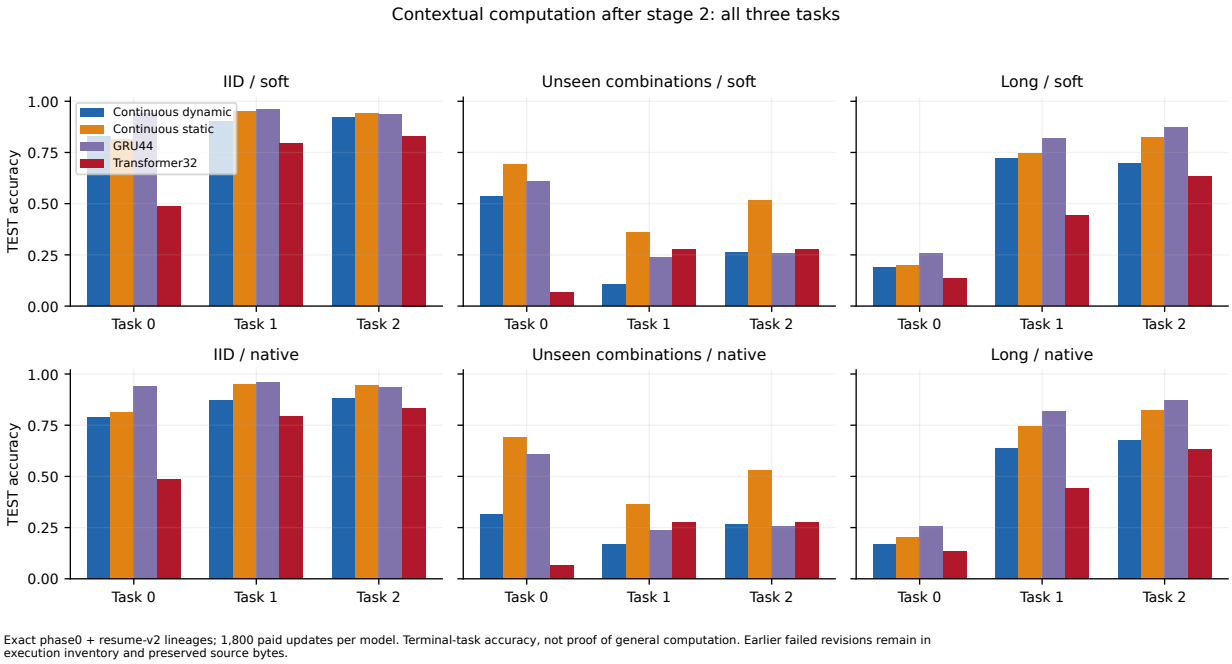


Figure 2: Three-task contextual development, separate IID, unseen-composition and long-sequence outcomes. Target skew and overwritten-read limitations apply.

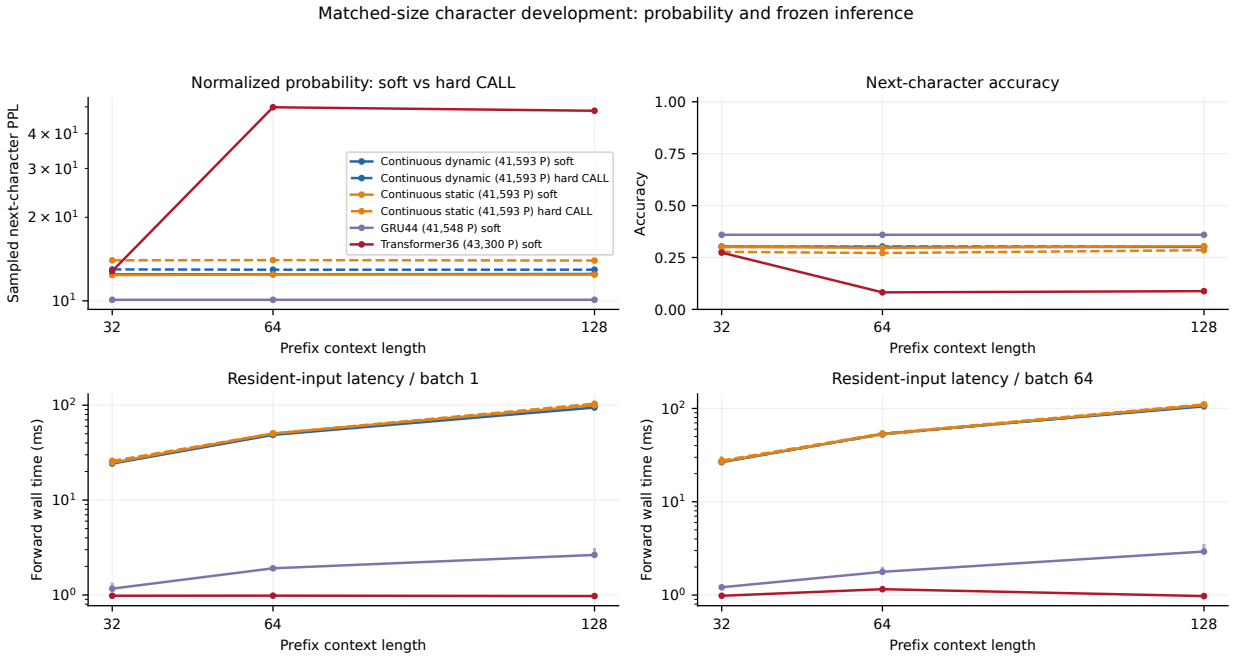


Figure 3: Earlier terminal-prefix character protocol: paired512 target positions and frozen serial inference. Hard CALL retains continuous probability readout. This figure is not pooled with the all-token protocol.

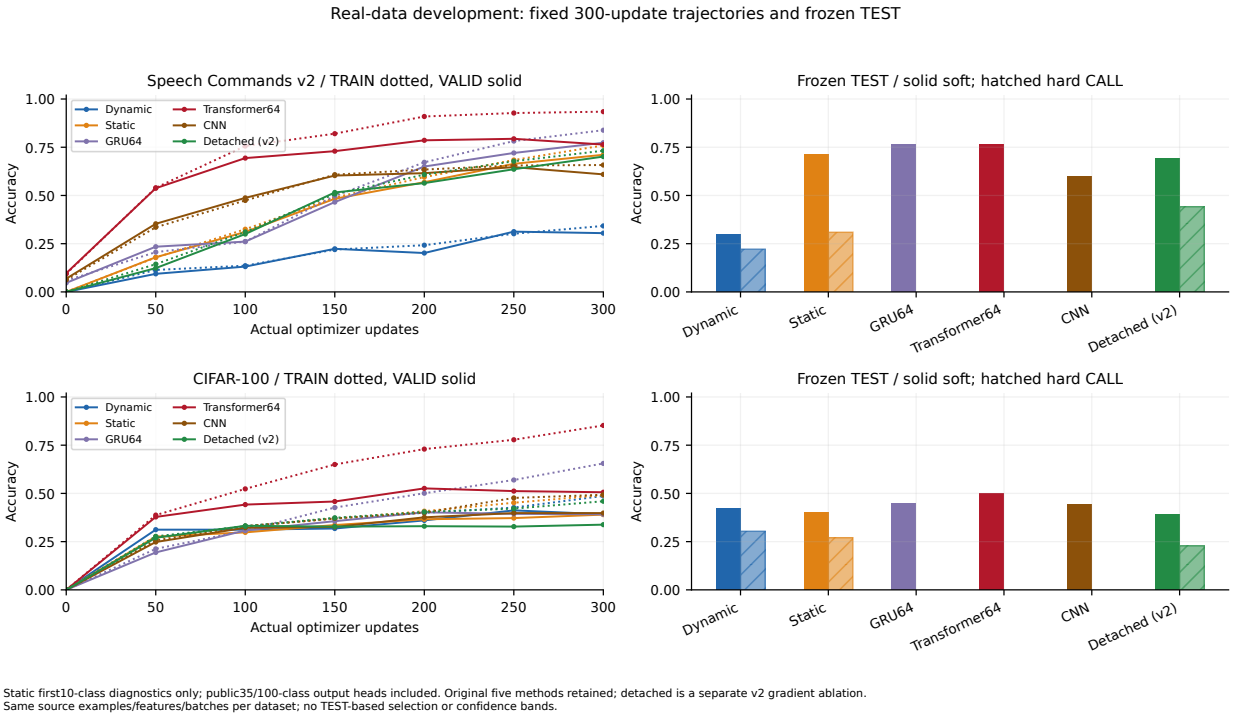


Figure 4: First-ten-class real-data development at16-dimensional Block interfaces: actual training/validation curves and frozen TEST results, including detached credit ablation. Not full-class or continual real-data validation.

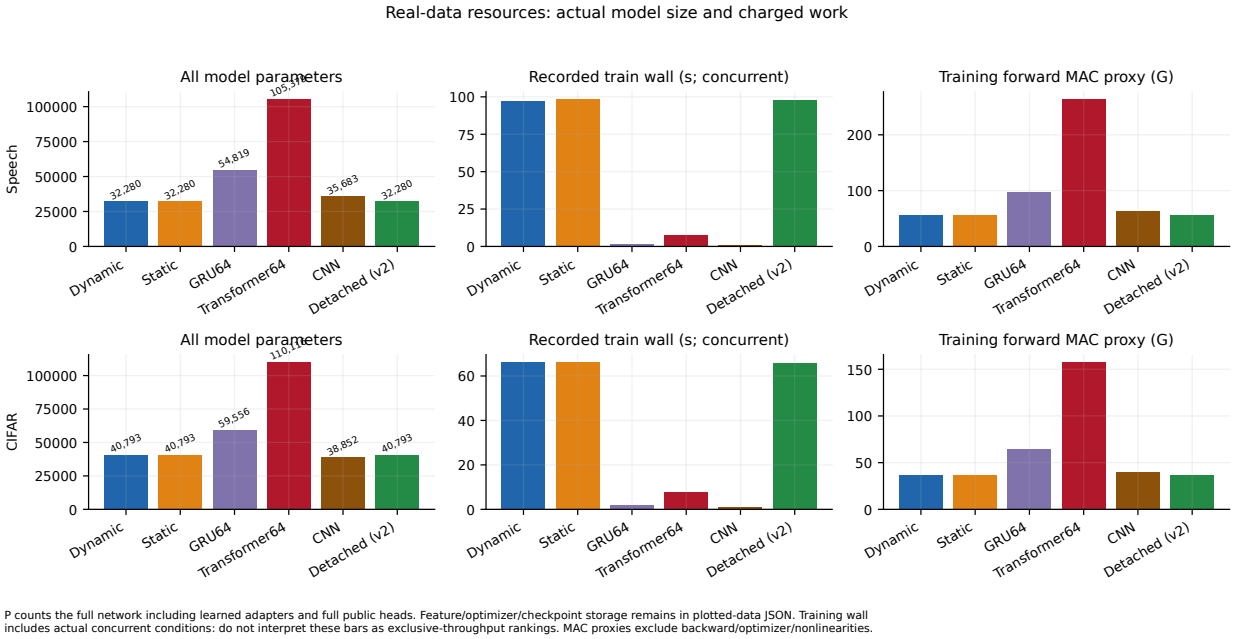


Figure 5: Parameters, actual concurrent training wall time, and forward-MAC proxies for the same first real-data development runs. Full storage/cost records remain separate; smaller MAC counts did not imply faster execution.

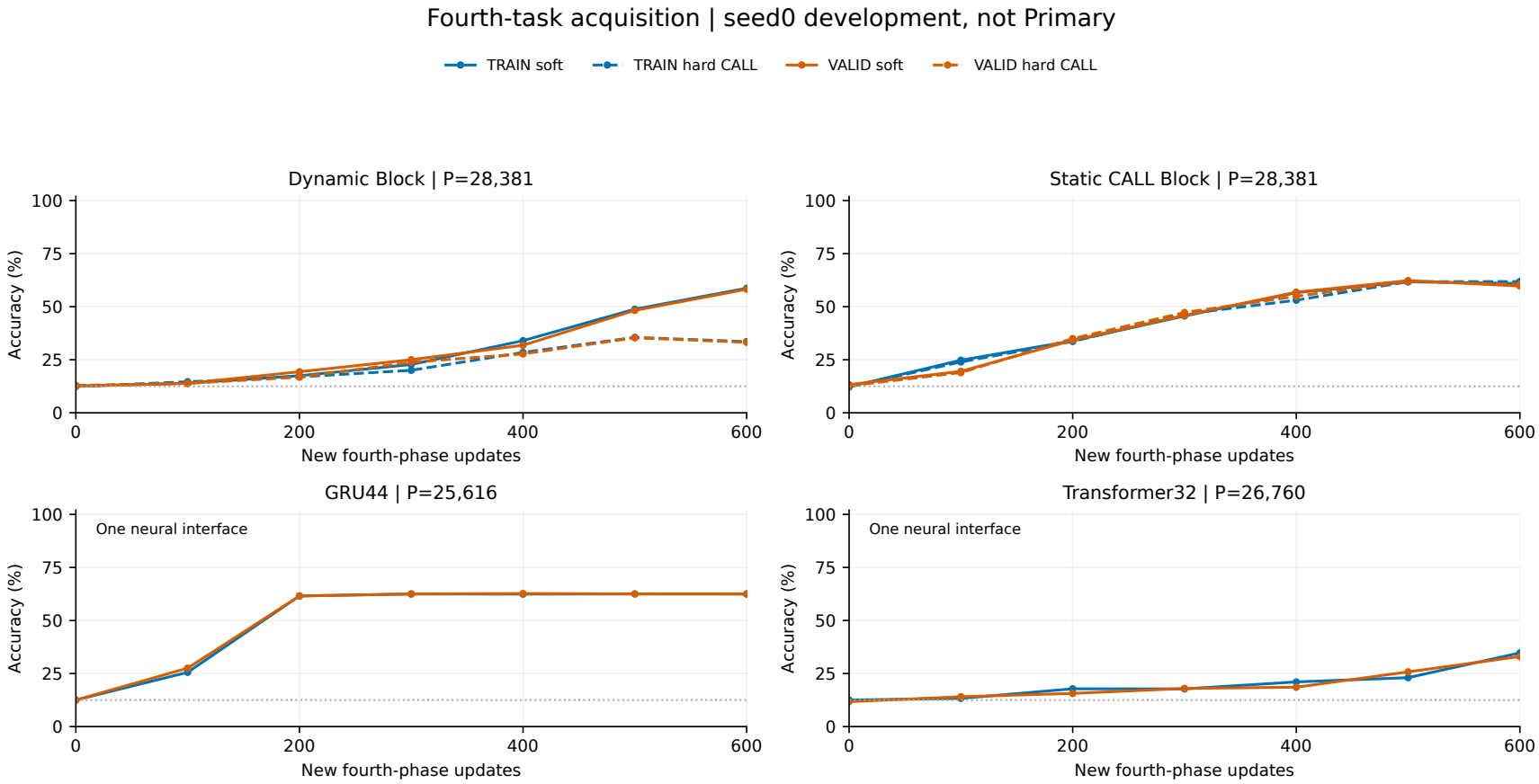


Figure 6: Fourth-task TRAIN/VALID acquisition: same 600 updates of 48 new plus 16 replay examples; no Primary claim.

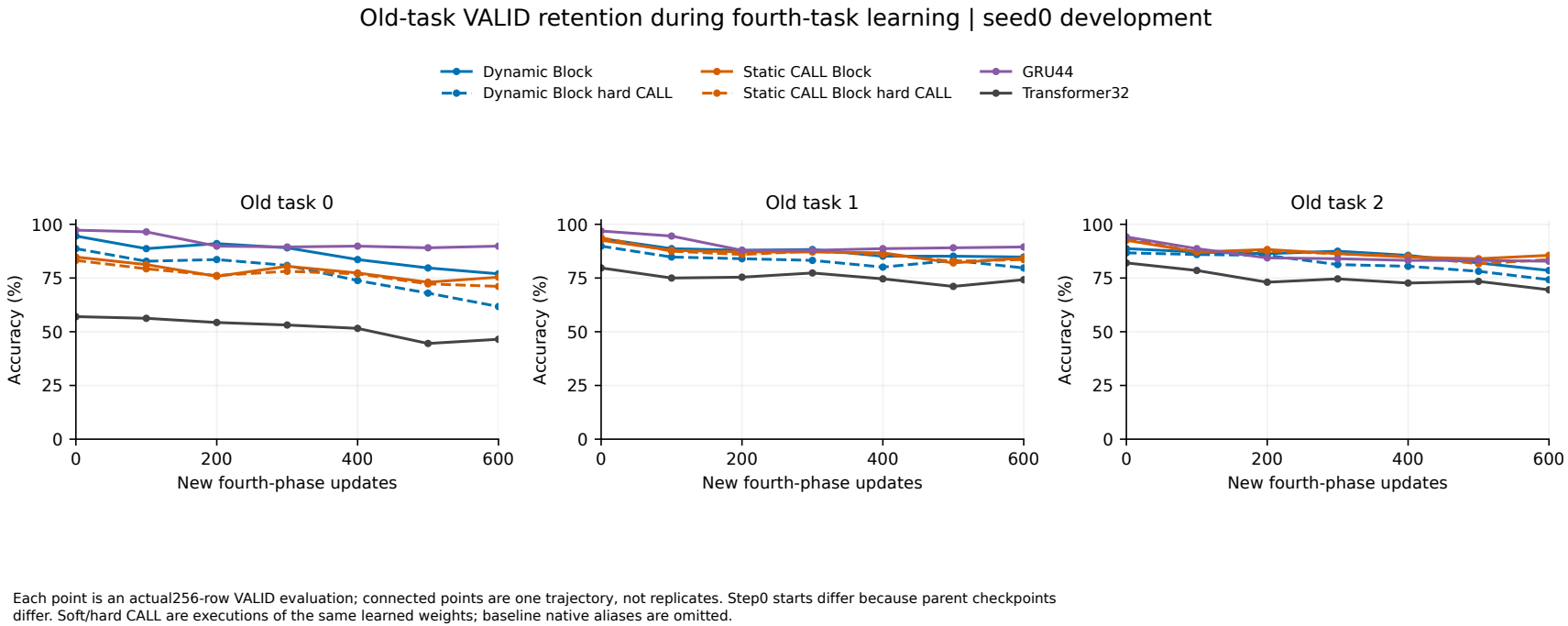


Figure 7: Old-task VALID retention: single trajectories, distinct actual parent checkpoints; no independent-seed confidence bands.

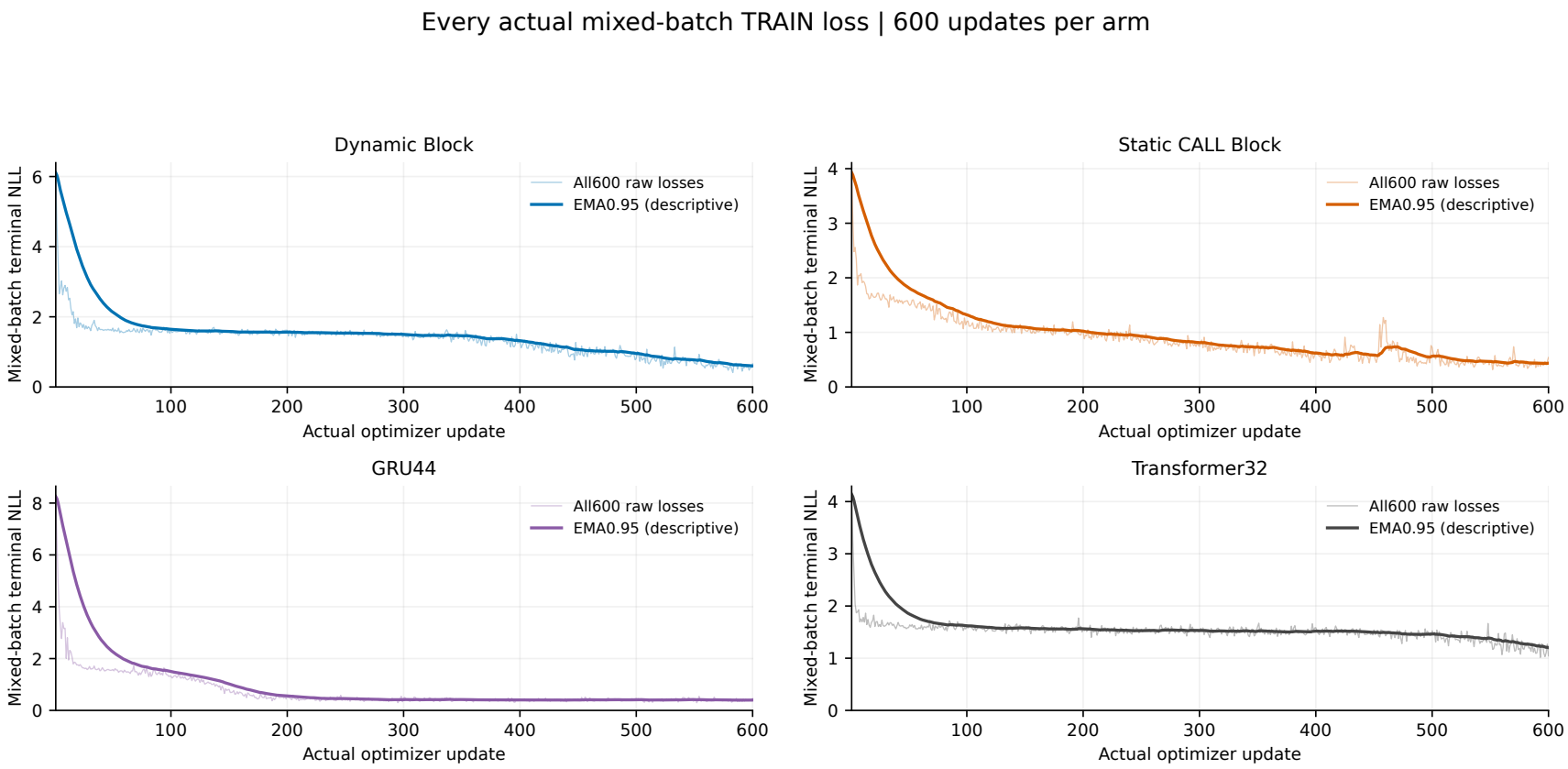
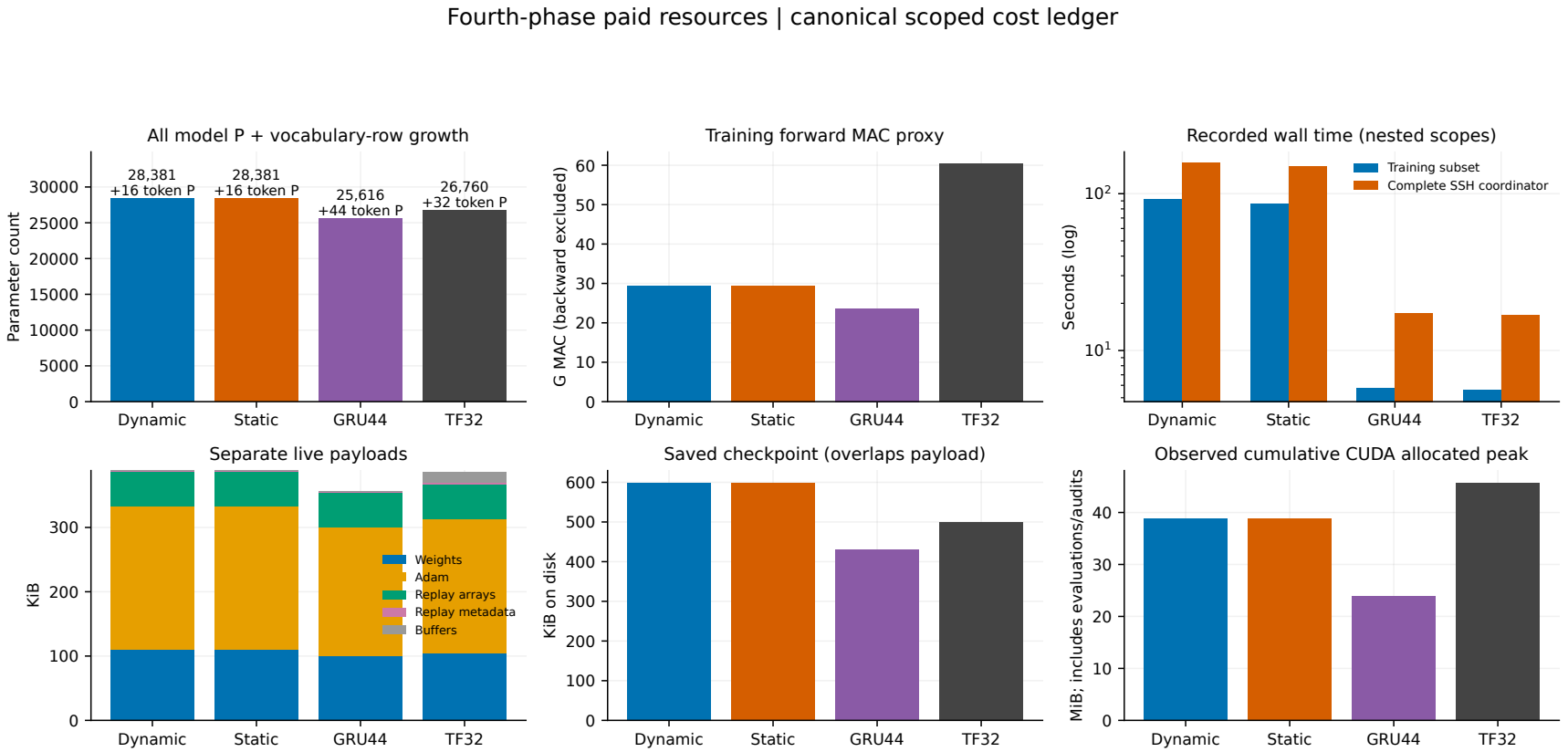


Figure 8: All 600 actual mixed-batch losses. The descriptive EMA is displayed with raw observations and was not used for selection.



All arms:600 updates,38400 label exposures=28800 new+9600 replay; Replay128 arrays=55296B. Added token P is NOT P per acquired/retained capability. SSH time includes training/evaluation/launch work; do not add nested fields or infer isolated throughput. Exact full training FLOPs/energy and capability-normalized delta P remain unknown; full scopes/CPU RSS/buffers retained in plotted-data and canonical ledger.

Figure 9: Joint parameter and scoped paid-resource comparison. Added vocabulary parameters do not establish parameters per retained new capability.

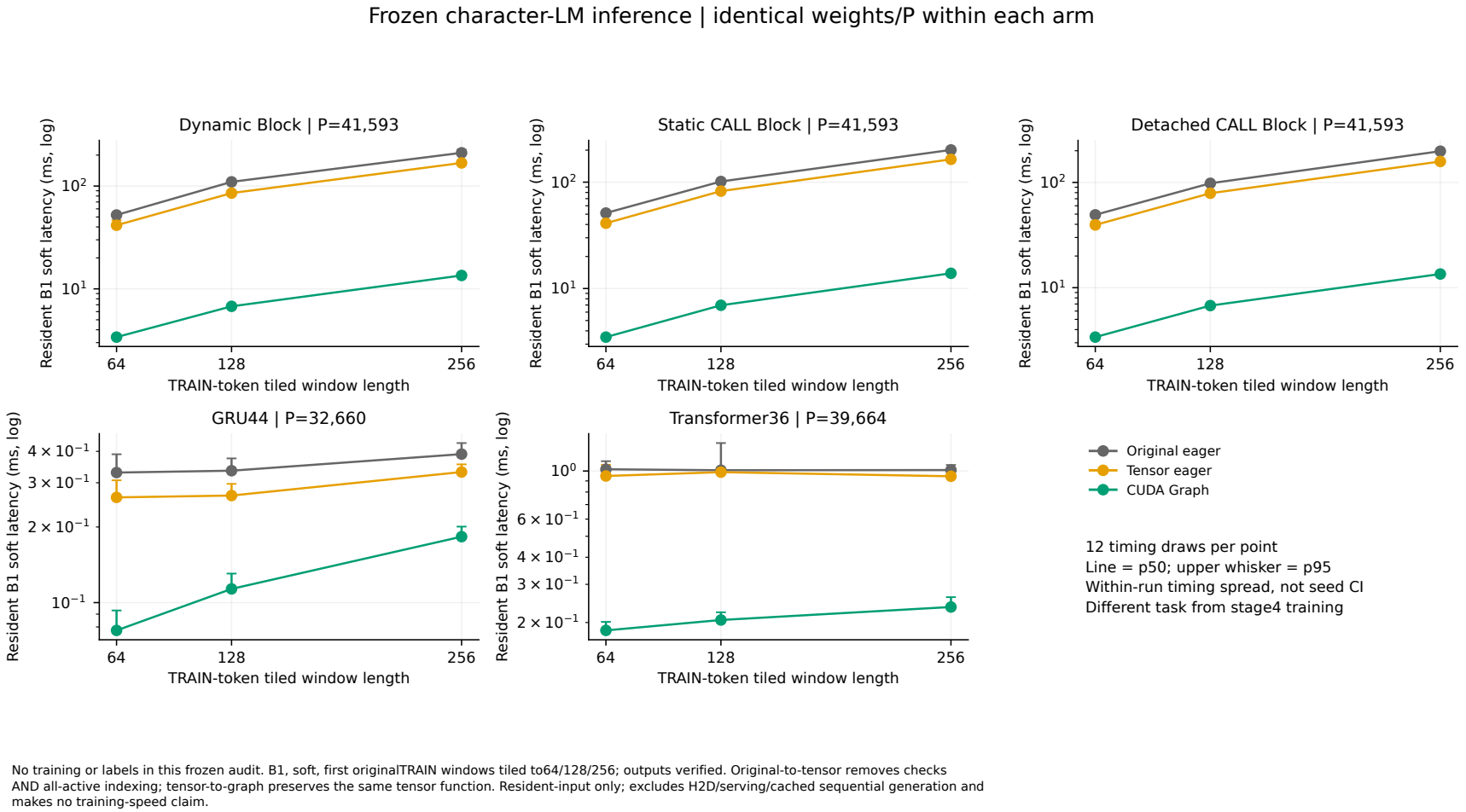


Figure 10: Same-weight character-LM execution costs, B1soft. Eager, tensor and CUDA Graph outputs were checked; conventional baselines remain faster.

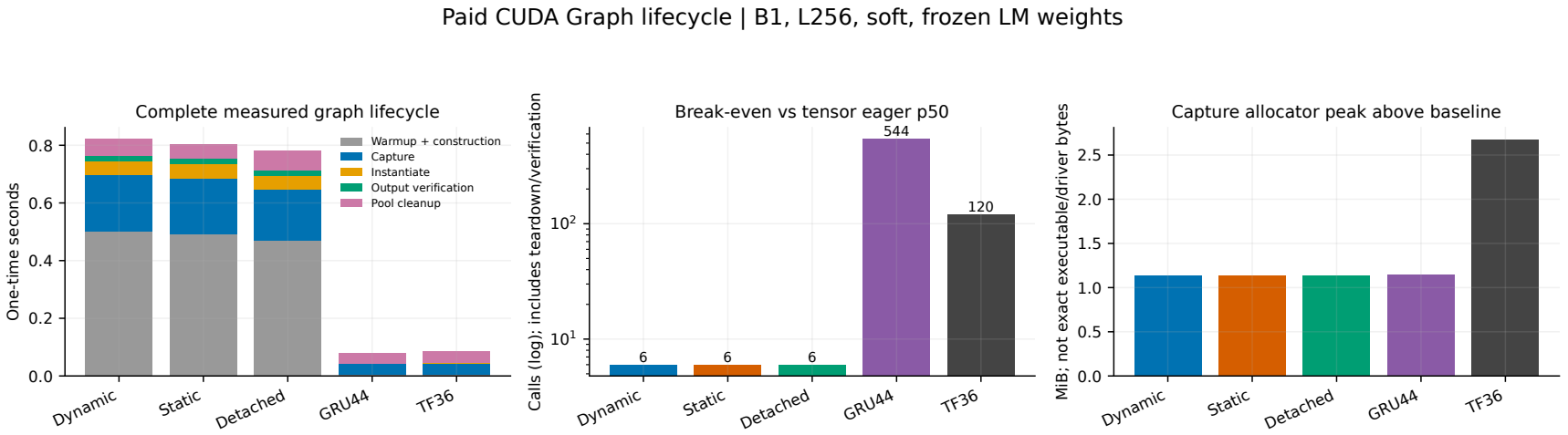


Figure 11: Graph capture, instantiation, verification and teardown costs, amortization and observed allocator changes. Exact driver executable bytes remain unavailable.

F Theory, evidence boundaries, and reproducibility

The role of theory is to identify the conditions under which an executable operation, its task binding, and its soft neural realization remain correct. These are distinct obligations. A mathematical invariant is not a proof of task acquisition, a finite successful training stream is not a generalization theorem, and an independent agent's agreement is not a scientific certificate. The results below describe conditional finite mechanisms; they do not assert an original general learning principle or a topology-specific cost advantage. Historical native/coordinate-renderer results and limited hybrid diagnostics remain separately versioned evidence. They are not measurements of Primary Prometheus, which is defined by the latest fully fused TODO 3-validated architecture and its subsequent frozen, actually retrained benchmark lineage.

F.1 Continuation witnesses and finite-state identification

Proposition 1 (Observed state separation)**.** *Fix a deterministic recurrent system and one task context. Suppose histories h and h' have different observed outputs under the same continuation s. Then any deterministic internal-state realization of both observations must assign different recurrent states to h and h'.*

Proof. If their recurrent states were equal, applying the same continuation under the same transition rule would produce equal final states and decoded outputs, contrary to the observations. □

Thus m pairwise distinct terminal continuation profiles require at least m distinguishable states in this deterministic representation. This justifies the historical prototype's witness-triggered workspace expansion. It does not show that equal finite profiles are equivalent under all unobserved suffixes. The implemented closure and full-TRAIN checks identify a model consistent with available observations; its held-out behavior remains an empirical question. No exponential global-closure conclusion is inferred from this finite witness.

F.2 Binding-level protection under state expansion

Let an old task model be (X, ι, T, r) and an updated model be (X', ι', T', r'), with a shared observable input domain and alphabet $\mathcal{A}$. The encoders are ι, ι', transitions are T_a, T'_a, and readouts are r, r'. A relation $\mathcal{R} \subseteq X \times X'$ protects the old computation if

$$(\iota(x), \iota'(x)) \in \mathcal{R} \quad \text{for every protected input } x, \tag{38}$$

$$(z, z') \in \mathcal{R} \implies r(z) = r'(z'), \tag{39}$$

$$(z, z') \in \mathcal{R} \implies (T_a(z), T'_a(z')) \in \mathcal{R} \quad (a \in \mathcal{A}). \tag{40}$$

Proposition 2 (Native execution preservation)**.** *Under these conditions, the old and updated systems have the same decoded output for every finite word over the protected alphabet, starting from any protected input.*

Proof. The empty-word state pair belongs to $\mathcal{R}$ by (38). Each next token preserves membership by (40). Induction on word length and readout agreement give the result. □

The relation can connect different-sized state spaces, so it permits expansion and internal reorganization without requiring every old parameter to freeze. It protects equivalence to the admitted reference, not correctness of that reference against an unknown task. Checking a finite reachable relation is ordinary semantic verification, not by itself a topological contribution. The reference construction, breadth-first exploration, acceptance/backtracking, and stored relation all have measured maintenance costs.

F.3 What continuous-loop protection establishes

For the present permutation-valued realization, a loop in the unordered configuration space of D distinct points induces a permutation of the marked fiber. With a fixed initial labeling, monodromy gives a representation

$$\rho : \pi_1(\mathrm{Conf}_D(\mathbb{C})/S_D) = B_D \longrightarrow S_D. \tag{41}$$

The quotient action records endpoint permutations; it is not a faithful representation of the whole braid group. Different braid classes, including pure braids, can have identical permutation actions. Ordinary graph equivalence is likewise not a new topological invariant merely because it is used alongside this realization.

Proposition 3 (A clearance condition for based-loop retention)**.** *Let $X = \mathbb{C} \setminus \mathcal{B}$ for a fixed finite branch set $\mathcal{B}$ and a fixed covering over X. Let γ be a polygonal based loop with clearance $\delta = \inf_{s,b\in\mathcal{B}} |\gamma(s) - b| > 0$. Let γ' have the same parameterized polygon structure and the same basepoint, and suppose the maximum corresponding vertex displacement is less than δ. Then γ and γ' are based-homotopic in X and induce the same marked-fiber monodromy.*

Proof. Linear interpolation on each segment bounds every pointwise loop displacement by the largest vertex displacement. For $H(t,s) = (1-t)\gamma(s) + t\gamma'(s)$, $|H(t,s) - b| \geq \delta - \sup_s |\gamma'(s) - \gamma(s)| > 0$. Thus the interpolation misses every branch point and fixes the endpoints. Homotopy invariance of covering-space lifting gives the same monodromy. □

This is a conditional geometric protection rule: it permits real continuous motion, rather than manually freezing all loop parameters. The covering, singular set, basepoint, and labeling must satisfy the stated fixed conditions; changing them requires its own argument. Applying it to a learned family also requires a bound for every protected loop/input in the claimed domain. Numerical continuation, finite-grid checks, and floating-point clearances are not automatically rigorous interval certificates. A certified deformation family may reuse an already paid-for realization across parameter changes, but its assumptions, initial lift, invalidation, and checkpoint reconstruction must all be included in the resource account.

Loop-class preservation, task-binding preservation, and accurate finite-time execution remain separate. A gate can remain topologically unchanged while a shared task router selects the wrong gate. Conversely, unchanged endpoint behavior under a conventional constrained program is not evidence that its learning benefit is topological. The software monodromy simulator is not a physical quantum computer; no hardware energy or quantum speedup follows from its numerical execution.

F.4 A sufficient soft/native consistency bound

Consider nonnegative stochastic soft execution as in (16). Suppose the hard computation uses at most $SL_{\max}$ operation slots. Let p_E, p_R, p_C, p_O be lower bounds on the probabilities assigned, respectively, to the selected native encoding, selected readout, every selected slot call, and every selected operation edge. The probability mass following that particular native execution path is at least

$$p_{\text{path}} \geq p_E p_R (p_C p_O)^{SL_{\max}}. \tag{42}$$

If this bound exceeds $1/2$, the native output is the unique soft argmax for every covered word, because all alternative output mass together is less than $1/2$. The same conclusion can hold with a weaker bound, since several paths may decode to the same output; this is only a sufficient condition.

The actual refresh evaluates these minima from the model's current soft operation tensors and its own hard calls. It checks parameter count, original slot structure, current native graphs, and frozen alias integrity. Subsequent weight or temperature changes require renewed checking. Floating-point evaluation of the analytical bound is not interval-certified verification. The result transfers current native correctness to a bounded-length soft interface; it does not teach a new class or prove always-online retention between maintenance points. The rule is available to the non-topological controls and therefore cannot establish a monodromy-exclusive effect.

F.5 Executed evidence and version lineage

Every reported run is tied to actual task arrays, configuration, code version, raw update, formation, and guard records, curves, saved neural tensors, extra state, and reload checks. Synchronization manifests supply file hashes for completed copies. A successful exit alone does not demonstrate learning; measured predictions and restored-checkpoint behavior supply the execution evidence. Missing records are retained as missing. A completed server-side artifact and a locally received runner exit receipt are distinct evidence: a transport timeout does not authorize inventing the latter. Failed code, unsuccessful model versions, and additional computation remain separate from the eventual successful trajectory, rather than being relabeled as if success occurred earlier.

The boundary evidence extraction reads original JSON/JSONL records and safe ZIP metadata without training or executing checkpoint code. Its CSV exposes stage-boundary outcomes and intermediate curves

separately. Subsequent additional seeds remain traceable to their own original run manifests rather than being silently attributed to that older extract. Frozen-backbone storage audits additionally check tensor aliasing and unchanged old weights; these audits establish stored values and counts, not independent neural inference. The raw GPU reload records provide the latter within the execution environment.

Inference reload is also distinct from exact training continuation. In the limited hybrid V2 diagnostic, actual neural weights and cold reexecuted maps were checked after reload, but the external Python smoothing RNG state was not saved. That checkpoint therefore establishes inference restoration, not bit-exact interrupted training recovery. Exact recovery requires the relevant RNG state, sampler/transaction position, and any optimizer or pending-update state, with an actual continuation check before making that stronger claim.

Reproducibility artifacts. The source, data generators, and raw run tree are under `.deep-work/2026-10-31T234403-0400-v2/probes/topology-architecture/`. The historical finite formation implementation is `terminal_profile_formation.py`, the historical compact model is `parameter_native_models.py`, the experiment driver is `parameter_efficiency_study.py`, and neural/native maintenance is `parameter_consistency_refresh.py` within that directory. Independent within-campaign examinations are preserved at `.deep-work/2026-10-31T234403-0400-v2/supervision/reviews/review-037-boundary-growth-and-frozen-readout.md` and `.deep-work/2026-10-31T234403-0400-v2/supervision/reviews/review-038-loop-boundary-learning.md`. The latter concerns a distinct, small terminal-feedback loop-class diagnostic; it does not substitute for the six-stage architecture or publication benchmark. The subsequent integrated hybrid V1/V2 evidence and its inference-versus-RNG recovery boundary are examined in `.deep-work/2026-10-31T234403-0400-v2/supervision/reviews/review-039-hybrid-loop-development.md`. That limited D3 development result is not full TODO 3 validation or a Primary architecture freeze.

The frozen growth extract's SHA256 is `311a5073e8f777fc6c6270d5323ebee258542b9be15a50d1bae2fa3ddb3e6c12`, and its learning-curve CSV has SHA256 `b7e9a6b9e11ce327e0e2b1d108ae3ea95b82e3f1483f32cc7e51d50f88af28b4`. These identify particular evidence snapshots, not a claim that every later artifact or current source has the same hash. Each executed version and later run must retain its own provenance.

F.6 Independent checking and limits of present claims

The campaign uses independent agent contexts to inspect original equations, code, raw outputs, costs, and saved tensor metadata. Such inspection has corrected coding, physical-model, cost, and claim-scope errors. It is not independent laboratory replication, community peer review, or proof by model consensus. The finite propositions above are mathematical arguments in the stated models; this manuscript does not claim Lean compilation or formal verification of their implementations. Primary-source reading records preserve what was actually read and the boundaries of adaptations. Historical novelty requires a checked comparison with substantive original work, beyond an independent reimplementation of the same rule.

The final reproducibility package requires fixed data splits, all five final seeds, locked configurations, executable environments, checkpoints, curves, raw failures, causal ablations, and statistical analysis scripts. Formal confidence intervals and paired comparisons remain to be produced under the protocol in Section E after the fully fused Primary's core, code, and configuration freeze. Historical architecture versions cannot be pooled into that lineage or selected as its substitute. Baseline configuration validity depends on implementation, claimed reproduction fidelity, fair information, and the prospective protocol, never on attaining an accuracy threshold. Low-scoring valid configurations and unsuccessful development trials remain in the evidence. Complete cost reporting must include development, discovery, formation, maintenance, storage, verification, and inference in addition to gradient updates.

At this checkpoint, historical evidence supports finite executable continual acquisition with boundary consistency and matched ordinary controls; the separate hybrid diagnostic supports a limited integrated learning repair. Neither supplies newly retrained Primary results. An independently useful topological learning contribution, a general nonreversible/continuous Block, competitive real-data results, and the complete publication-grade evidence package remain unresolved. This manuscript is a working scientific record, not a declaration of breakthrough, publication readiness, or public release; authorship and intellectual-property conditions must be settled before external dissemination.

F.7 Training-state recovery: executed new audit

The original hybrid V1/V2 checkpoints support inference reload but omit their external smoothing RNG state. That missing history remains missing. A new checkpoint wrapper additionally records the Python sampling generator, Torch CPU RNG, tensor bits and strides, parameter gradient flags, training control, source hashes, semantic receipts, cache state, and logical work counters.

A separately seeded audit starts from the actual V2 transaction-30 checkpoint, performs one new gradient calculation, saves, and compares the next continuous step against the next step after disk reload. The three actual new gradient calculations match in all ten frozen comparison groups, including both credit components, complete gradients, neural tensors, RNG, execution outputs, semantic state, and work counts. Physical restore/constructor overhead is recorded separately from restored logical training history. This is a CPU explicit-SGD recovery check, not reconstructed original V2 training or a CUDA guarantee.

The first audit attempt failed before any gradient calculation because the isolated process could not read a checkpoint placed beside its inspected source files. The retry places the identical input in the runner's existing read-only input allowance; isolation and source rules are unchanged. Both attempts and their actual exit receipts are retained under `probes/topology-architecture/runs/parameter-hybrid-training-recovery-v1/` and `parameter-hybrid-training-recovery-v2/` in the same campaign.